

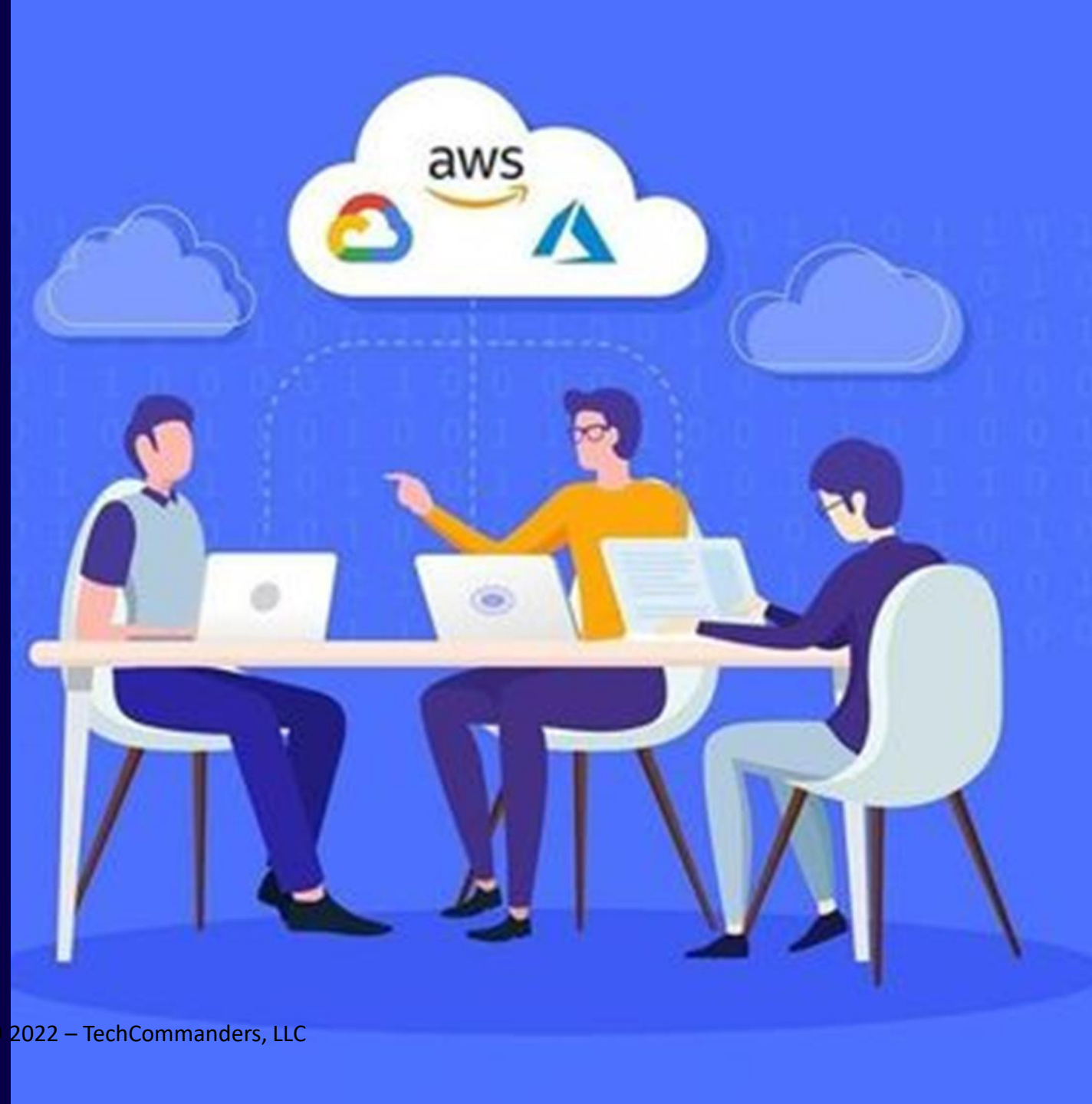
Google Cloud FinOps Fundamentals

Understanding FinOps Management on Google Cloud



Introduction

- Course Introduction
- Course Flow
- Instructor Information



Google Cloud FinOps Fundamentals

SECTION 1 – FINOPS
FUNDAMENTALS

SECTION 2 – GOOGLE
CLOUD TOOLS AND
SERVICES FOR
FINOPS

SECTION 3 -
MANAGING SERVICE
COSTS

COURSE CLOSEOUT

Course Expectations

- Whiteboard Discussions
- GCP Demos
- Presentation download
- Documentation Links



About Your Instructor

Joseph Holbrook, CLO of Techcommanders in Jacksonville, FL

- Certified Blockchain Solutions Architect (CBSA)
- Certified Google Cloud Platform Cloud Architect
- Certified AWS Solutions Architect
- FinOps Practitioner
- Brocade Distinguished Architect (BDA) 2013
- EMC Proven Professional – Expert – Cloud (EMCCE)
- Published Course Author on Pearson Safari, Udemy, LinkedIn Learning
- Published Book Author – Architecting Enterprise Blockchain Solutions
- CompTIA Subject Matter Expert, SME
- Prior US Navy Veteran



Course Pre-Requirements

Let's confirm your ready for the course.

Google Cloud FinOps Fundamentals

Pre-Requirements

- 6 Months of active use with Google Cloud
- Free Tier or Paid Account for practice
- Download Materials for Review



Section 1

FinOps Fundamentals

What is FinOps and getting to know the basics.

Section Overview

WHAT IS
FINOPS?

Why FinOps is
important?

FinOps
Culture

FinOps
Principles

FinOps Teams

FinOps
Lifecycle and
Phases

FinOps
Triangle

Best Practices

Aligning
Teams and
Resources

Whiteboard -
Putting it all
together

Section
Review

Section
Review
Questions

What is FinOps

Let's Define and Identify what this really means.

What is FinOps about?

- The practice of bringing financial accountability to the variable spend model of cloud, enabling teams to make business trade-offs between spend, cost and quality.

-FinOps Foundation



Google Cloud FinOps Fundamentals

What is FinOps about?

- FinOps is a collaborative effort between Finance and Operations where cloud spending is moderated and optimized.
- AKA – Cloud Financial Management
- A **culture** to adopt
- Method of ROI and cost optimization





What is FinOps about?

- An efficient approach for teams to manage their cloud costs.
- Ownership driven and supported by a central group
- Work towards financial and operational control.
- A way to prioritize ongoing optimization of your cloud costs.

Google Cloud FinOps Fundamentals

FinOps should provide Results!

- Companies must be able to take advantage of “variable spend” model of Cloud
- A centralized team should provide value and drive FinOps.
- Reporting must timely and actionable with incentives
- Ownership for everyone in the enterprise.
- Realize business value to drive decisions
- Crawl, Walk and Run



Why use a FinOps Approach

Does Cloud Really need FinOps?

Google Cloud FinOps Fundamentals

Why use FinOps Approach?



- Cloud Spending increasing YoY due to “variable spend” of cloud.
- Managing Cloud Spending can be overwhelming w/o a culture shift
- Set of Best Practices
- Provide corporate oversight of cloud spending. (Centralized)
- Provide for a solid ROI
- Forecasting Accuracy (Actionable)

Culture of FinOps

Organizational Needs to Adapt

Google Cloud FinOps Fundamentals

Establishing a FinOps Culture

An organization's FinOps culture embraces a long-term roadmap of transformation and continuous improvement across all the FinOps Domains at each stage of FinOps Maturity. (FinOps Foundation)

- Cultural Practice (proactive)
- Taking Ownership
- Collaborative
- Best Practices



Principles of FinOps

Six Principles to Understand

Google Cloud FinOps Fundamentals

Six Principles of FinOps

1. Collaboration
2. Business Value
3. Ownership
4. Timely Reports
5. Centralized Team
6. Variable Cost Model



Google Cloud FinOps Fundamentals

FinOps is a process and a set of best practices to bring the following aspects to companies and stakeholders:

- Collaboration – FinOps is not about one person at a company but a constant collaboration between engineers and their managers, between R&D, Operations and Financial departments, CTO, CIO and VPs offices.
- Business Value – FinOps needs to be implemented to provide value for the organization. Provide for cost optimization and a solid ROI
- Ownership - Stakeholders need to be able to control and take ownership of their expenses.
- Timely Reports – Reports should be routine and concise to provide insight into cloud spend and forecasting.
- Centralized Team - A FinOps team should be centralized to provide for an enterprise focused approach removing redundancy but provide for collaboration.
- Variable Cost Model – Cloud Spending is variable and therefore the FinOps team must have processes, practices and tools in place to manage and monitor spending to meet the business goals.

FinOps Lifecycle and Phases

There's a documented journey with FinOps

Google Cloud FinOps Fundamentals

What is the FinOps Lifecycle?

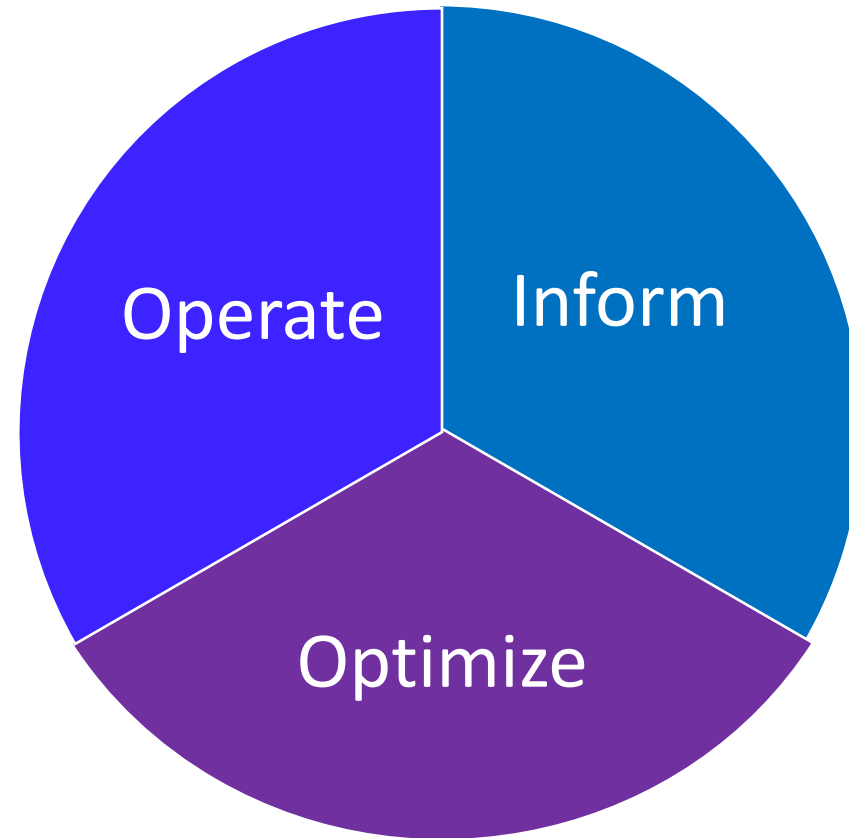
“The FinOps Lifecycle surrounds the idea of supporting a Culture of Accountability and a Governance framework to support the decision making FinOps makes possible”

- Inform
- Optimize
- Operate



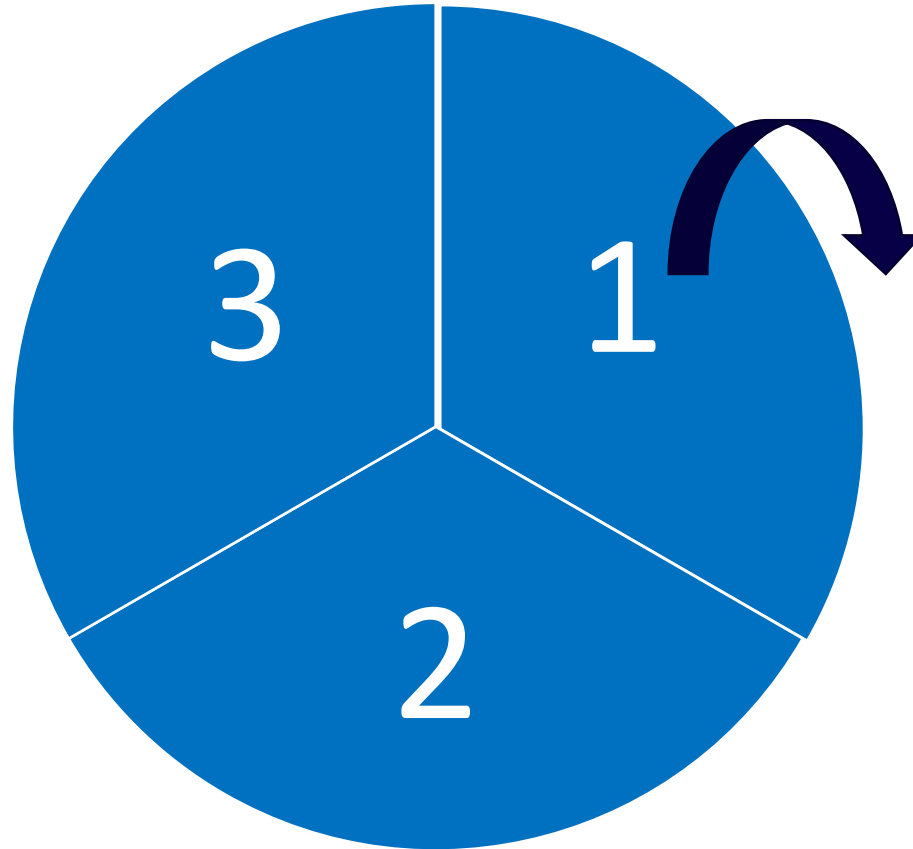
Google Cloud FinOps Fundamentals

Phases of the FinOps Lifecycle



Google Cloud FinOps Fundamentals

Phases of the FinOps Lifecycle



Google Cloud FinOps Fundamentals

INFORM

- Understand your organizations resource consumption thru context, data, budgets and Forecasting.(Visibility and Allocation)

OPTIMIZE

- Setting your goals and targets by determining you cloud costs thru strategy, tracking, cost breakdown and cost growth boundary.(Utilization)

OPERATE

- Aligning organization teams to business goals thru drive and action. Focus is on getting results for the organization. (CI)

Google Cloud FinOps Fundamentals

Lifecycle Phase	Main Tasks/Goals	Principles Supported
Inform	- Create tags/labels - Setup account hierarchy - Goal is for near real time cost visibility	Set Goals
Optimize	- Set the goals and determine the impact. - Achieve the goals	Principle 1: Teams need to collaborate; Principle 2: Business value of cloud drives decisions; Principle 6: Take advantage of the variable cost model of the cloud
Operate	- Target, define, and document optimization opportunities - Automation	Principle 2: Business value of cloud drives decisions Principle 5: A centralized team drives FinOps Principle 6: Take advantage of the variable cost model of the cloud

Google Cloud FinOps Fundamentals

Purpose of the Phase	Google Cloud Epics
Inform – Provide Transparent, accurate and actionable data	- Allocate (labels, tags, metadata) - Reporting KPI - Reporting Costs - Forecasting
Optimize – Create efficiency in the organization with controls	- Product Efficiency - Pricing Efficiency - Guardrails
Operate – Maintain the collaboration, constant innovation with incentivizing and measurements	- Onboard Workloads - Educate - Incentivize - Measure Value - Funding

What is a FinOps Team

Understanding Teams

Google Cloud FinOps Fundamentals

- FinOps effectively brings Finance, Tech and Business teams together to improve business efficiency while bringing value by using cloud technologies.
- The art of FinOps is to bring this combination together with a change in team culture and internal processes.
- Collaborative and Centralized



Google Cloud FinOps Fundamentals

The FinOps team should be responsible for:

- Defining enterprise cloud usage strategy
- Defining and adjusting enterprise cloud budgets
- Setting up the cloud usage practices and guidelines
- Reviewing the data and determining corrective actions as needed

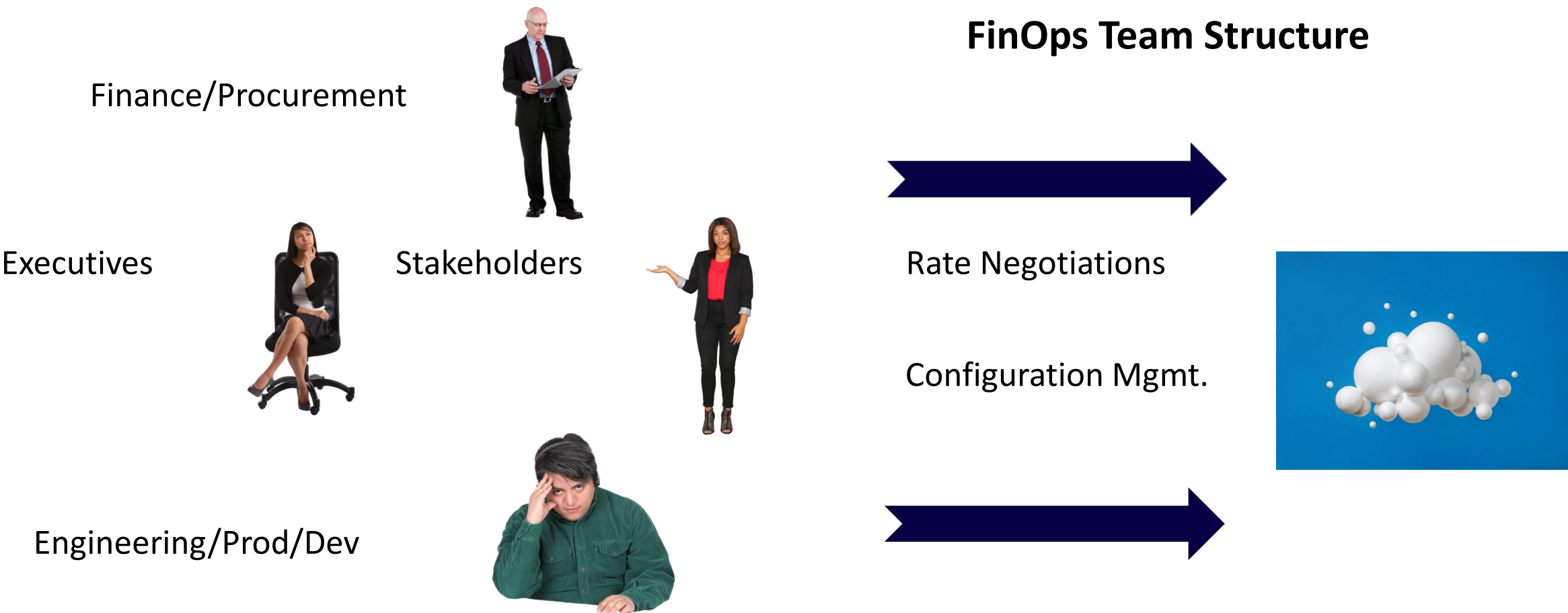


Google Cloud FinOps Fundamentals

- Cloud costs need to be visible to everyone.
- Finance should not be the only BU managing costs
- In the cloud we use a shared responsibility model.



Google Cloud FinOps Fundamentals



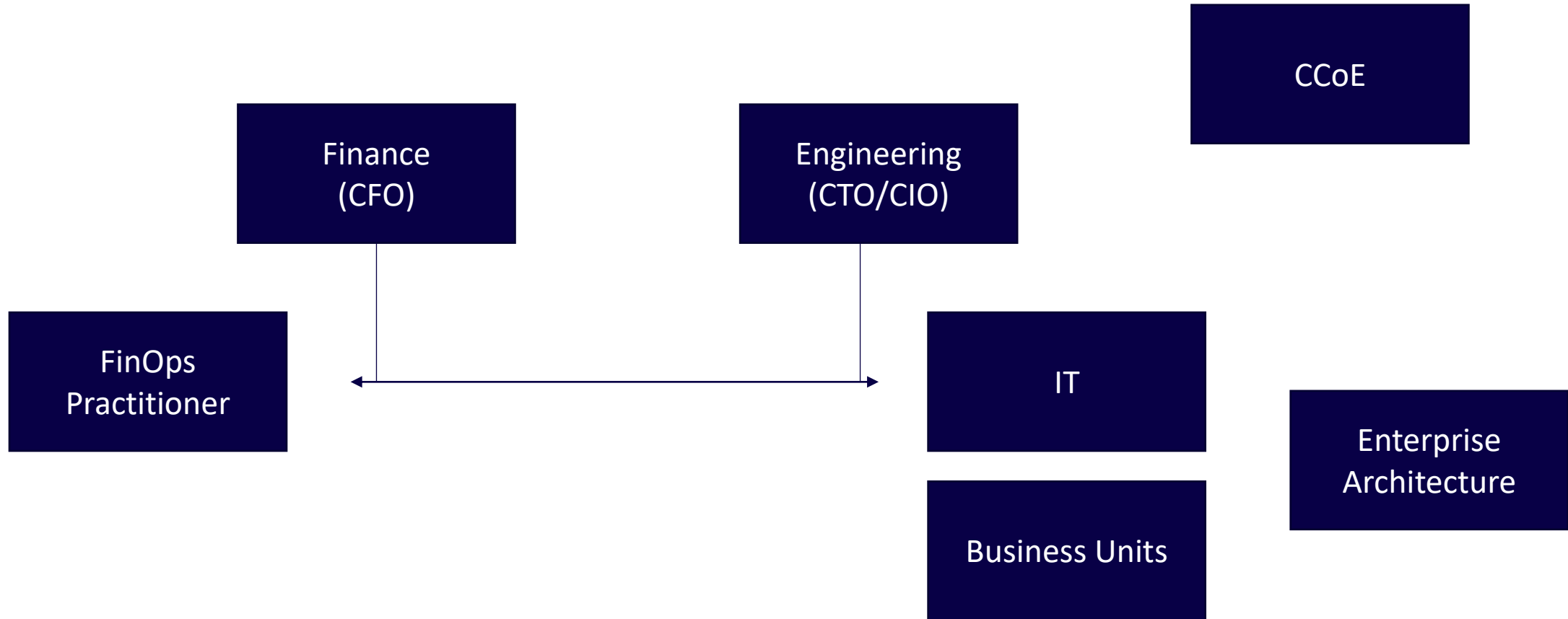
Google Cloud FinOps Fundamentals

FinOps Team structure should be considered for the enterprise

- Stakeholders
- Centralization
- Meet Business Goals
- Roles and Responsibility



Google Cloud FinOps Fundamentals





Google Cloud FinOps Fundamentals

Establishing a team by identifying specific members from:

- Cloud Center of Excellence (CCoE)
- Sponsor(s)
- Engineering
- Finance
- Application Owners
- DevOps
- And more business units as needed

Google Cloud FinOps Fundamentals

FinOps Team Should be placed

- FinOps Team should be under the CFO/COO in most enterprises.
- Should be aligned with engineering/development
- Should be collaborative with Finance and Engineering to meet business goals.



Google Cloud FinOps Fundamentals

Evaluating Reserved Instances (GCP – Committed Use Discounts)

- A Net Present Value (NPV) analysis could be used for evaluating Reserved Instances



FinOps Triangle

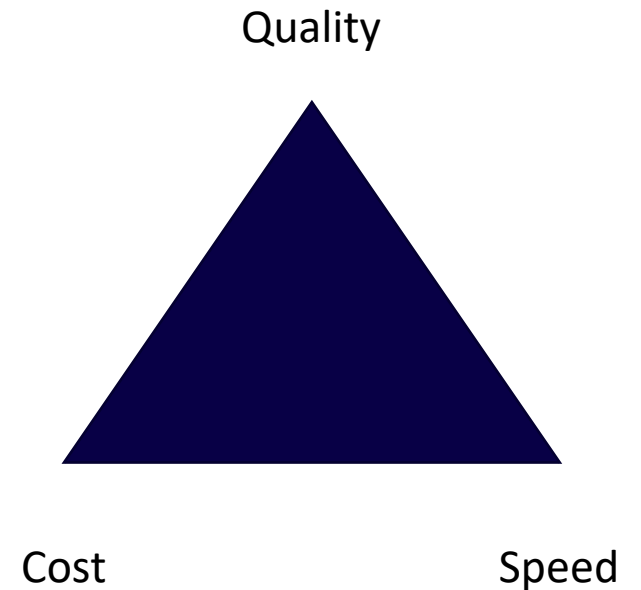
Trade - Offs

Google Cloud FinOps Fundamentals

The FinOps Iron Triangle is like the project triangle

Trade Offs (Compromise) – Quality, Speed and Cost

- Speed: How fast we would like to deliver.
- Quality: Number of bugs, performance, ease of use, etc...
- Cost: Total cost for development and maintenance



Best Practices

Best Practices

Google Cloud FinOps Fundamentals



Inform - Allocate, benchmark, and forecast cloud service needs.



Optimize –Compare the needs of the company with discounts for time and reserved usage in order to get the better rates.



Operate - Continuously evaluate the performance of cloud services against business objectives for speed, quality and cost.

Google Cloud FinOps Fundamentals

Best Practices

- Cloud stakeholders need to agree on a comprehensive organizational grouping and tagging regime.
- Cloud as a business value generator
- Build Communication channels for collaboration (remove silos)
- Develop a common vocabulary
- Centralize a FinOps organization.
- Setup Kubernetes shared cost allocation models.

Google Cloud FinOps Fundamentals

Best Practices

- Forecasts for a 3–12-month timeframe.
- Develop forecasts with the help of machine learning tools.
- Develop scorecards aimed at comparing teams.
- Use Real Time Reports
- Expand Reporting and Forecasting
- Shift Left Cost Accountability
- Understand Cost Influence - Cost avoidance and Rate reduction.

Aligning Teams to Resources

Ensuring the right people manage the right resources

Google Cloud FinOps Fundamentals

Managing People and Resources

- Resources – VMs, Services, Storage, etc
- Teams – Business Units, Define stakeholders, structure, train



Google Cloud FinOps Fundamentals

Teams (people)

- Executive buy-in for FinOps and its practices and processes is required.
- Teams need to collaborate
- Continuously improve for efficiency and innovation.
- Everyone takes ownership of their cloud usage.
- Shift Left
- Technical teams balance performance and costs
- Engineers and operations teams stay focused on usage optimization.

Google Cloud FinOps Fundamentals

Resources

- Enhance visibility into resource usage
- Instance types – Reserved, On Demand, Spot, etc.
- Volume/Custom Discounts with cloud providers

Whiteboard – Considerations Summary

Discuss Lifecycle

Google Cloud FinOps Fundamentals

Putting it all Together



Section Summary

Section 1 : FinOps Fundamentals



1

Section

Section Summary

- FinOps is a collaborative effort between Finance and Operations where cloud spending is moderated and optimized.
- FinOps is meant to help your enterprise to prioritize ongoing optimization of your cloud costs and reduce waste.
- “The FinOps Lifecycle surrounds the idea of supporting a Culture of Accountability and a Governance framework to support the decision making FinOps makes possible
- Inform - Allocate, benchmark, and forecast cloud service needs.
- Optimize –Compare the needs of the company with discounts for time and reserved usage in order to get the better rates.
- Operate - Continuously evaluate the performance of cloud services against business objectives for speed, quality and cost.
- FinOps Triangle is like the project triangle in the sense that if we adjust one Trade Offs (Compromise) – Quality, Speed and Cost that it would affect the other two Trade Offs

Section Review Questions

Section 1 : FinOps Fundamentals



Review Questions

The FinOps Lifecycle has three phases. What are they? (Select Three)

- Inform
- Exploit
- Optimize
- Evolve
- Operate

Review Questions

The FinOps Lifecycle has three phases. What are they? (Select Three)

- Inform
- Exploit
- Optimize
- Evolve
- Operate

Review Questions

In what phase of the FinOps Lifecycle do we create efficiency in the organization with cloud use and budget controls?

- Inform
- Optimize
- Operate

Review Questions

In what phase of the FinOps Lifecycle do we create efficiency in the organization with cloud use and budget controls?

- Inform
- **Optimize**
- Operate

Review Questions

What business unit in the enterprise should the FinOps Team be reporting under?

- CTO
- CIO
- CFO
- DevOps
- Production

Review Questions

What business unit in the enterprise should the FinOps Team be reporting under?

- CTO
- CIO
- CFO
- DevOps
- Production

Review Questions

Which of the following answers would be correct around Cloud Computing and how resources are consumed? (Select Two)

- Cloud computing introduces new ways to consume and pay for IT services
- The shift from CAPEX to OPEX financing and cash flow models using cloud
- The shift from OPEX to CAPEX financing and cash flow models using cloud
- Cloud computing introduces new culture and new cash flow methods

Review Questions

Which of the following answers would be correct around Cloud Computing and how resources are consumed? (Select Two)

- Cloud computing introduces new ways to consume and pay for IT services
- The shift from CAPEX to OPEX financing and cash flow models using cloud
- The shift from OPEX to CAPEX financing and cash flow models using cloud
- Cloud computing introduces new culture and new cash flow methods

Section 2

Google Cloud Tools/Services FinOps Focused

Getting to know how to manage FinOps with Google Cloud

Section Overview

FINOPS TOOLS/ SERVICES ON GCP	FinOps Epics (Google)	Google Cloud Services Cost Optimization	Google Cloud Resource Management	Cloud Infrastructure Selection
Orgs and Projects	Demo - Organizing your Resources	Demo - Billing	Demo - Invoices	Demo - Reporting
Tag/Label Management	Demo - Tag/Label Management	Cloud Pricing Calculator	Demo - Pricing Calculator	Whiteboard - Putting it all together
Section Review		Section Review Questions		

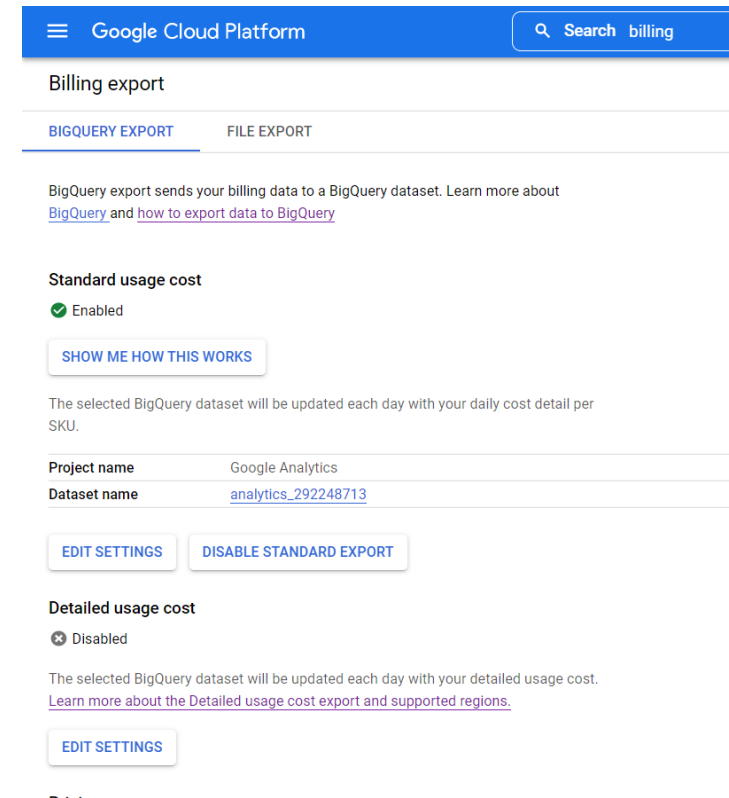
FinOps Tools and Services

What is available with GCP?

Google Cloud FinOps Fundamentals

Tools/Services available in GCP that aid with FinOps (Cost Management Tools)

1. Billing Exports (BigQuery) and API's
2. Cloud Hierarchy
3. Labels and Tags
4. Reports

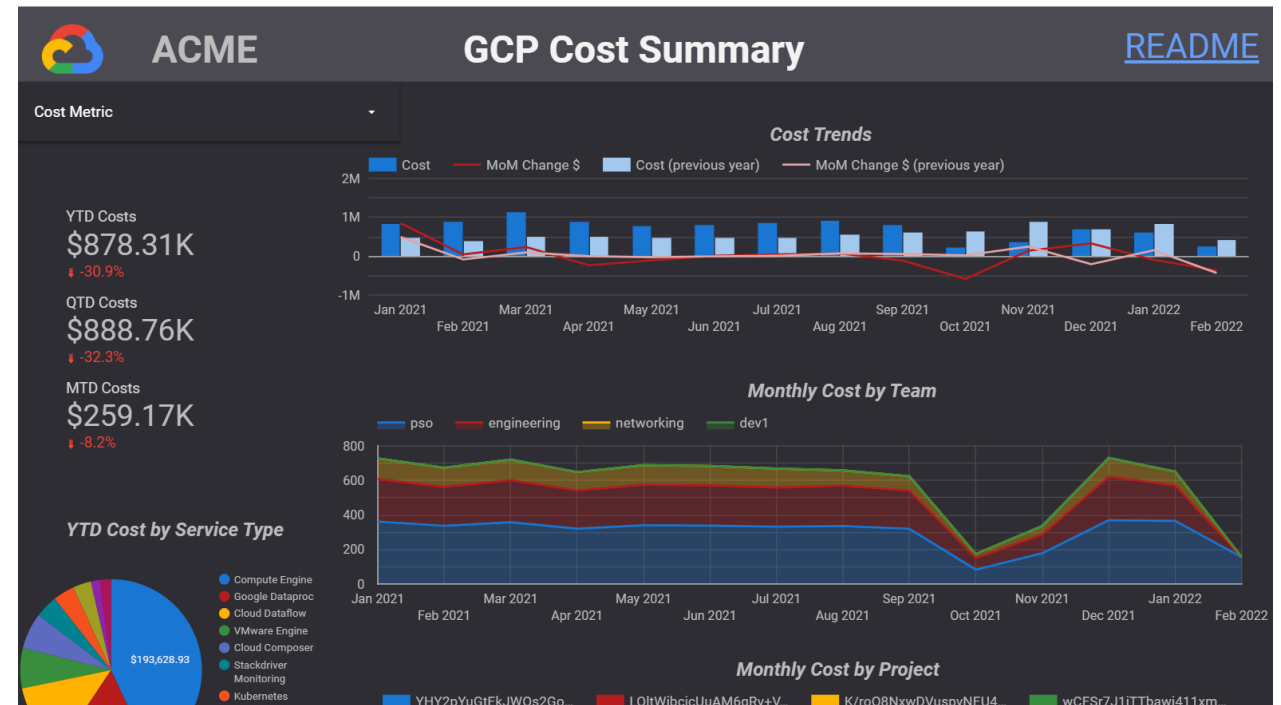


The screenshot shows the 'Billing export' settings in the Google Cloud Platform. The 'BIGQUERY EXPORT' tab is selected. It indicates that the 'Standard usage cost' export is 'Enabled' with a green checkmark. A 'SHOW ME HOW THIS WORKS' button is present. Below this, it states that the selected BigQuery dataset will be updated daily with daily cost details per SKU. The 'Project name' is 'Google Analytics' and the 'Dataset name' is 'analytics_292248713'. There are 'EDIT SETTINGS' and 'DISABLE STANDARD EXPORT' buttons. The 'Detailed usage cost' export is shown as 'Disabled' with a greyed-out button and a crossed-out icon. It also mentions daily updates and provides a link to learn more about detailed usage cost exports.

Google Cloud FinOps Fundamentals

Tools/Services available in GCP that aid with FinOps

1. Dashboards (Data Studio)
2. Cloud Operations
3. Recommendations
4. Quotas



Google Cloud FinOps Fundamentals

Budgets & alerts

+

CREATE BUDGET

🗑️

DELETE

Budgets track expenses within a Google Cloud Platform project or billing account. Your budget can be a specified amount or based on previous spend. You can set alerts to notify billing admins and users when a budget goes over a specified amount.

?

Setting a budget does not cap resource or API consumption. [Learn more.](#)

Filter

Enter property name or value

?

☐	Budget name ↑	Budget period	Budget type	Applies to	Trigger alerts at	Spend and budget amount
☐	FinOps Notification	Monthly	Specified amount	This billing account	50%, 90%, and 100%	\$45.69 / \$1,000.00 Includes -\$22.94 credit
☐	Monthly Alerts	Monthly	Specified amount	This billing account	50%, 90%, and 100%	\$45.69 / \$100.00 Includes -\$22.94 credit

RECENT

STARRED

ALL

	Name	ID
☆ 🔗	Google Analytics ?	encoded-hangout-331014
☆ 🔗	GCPDemoproject ?	gcpprojectdemo1234
☆ 🔗	GKEClass ?	gkeclass1

Cloud Billing Budget API

Cloud Billing Budget API Overview

A general overview of the Cloud Billing Budget API.

Cloud Billing Budget API Prerequisites

Cloud Billing concepts and other information you should know before you start coding your first Cloud Billing budget application.

Cloud Billing Budget API Setup

Configure the API and authorization to get ready to start using the Cloud Billing Budget API.

Using the Cloud Billing Budget API

Tutorial to create small but functional basic client applications using the Cloud Billing Budget API.

Cloud Billing Budget REST API

View, create, and manage budgets programmatically and associate them with products and projects.

Cloud Billing Budget RPC API

View, create, and manage budgets programmatically and associate them with products and projects.

Google Cloud FinOps Epics

Let's cover some of the Google Cloud FinOps Epics

Google Cloud FinOps Fundamentals

“When considering your next steps toward FinOps, it’s important to remember that an iterative process is needed. No organization can go from zero to fully realized FinOps in a short period of time.”

– Google Cloud FinOps Whitepaper

Google Cloud FinOps Fundamentals



EPICS AND PROCESSES

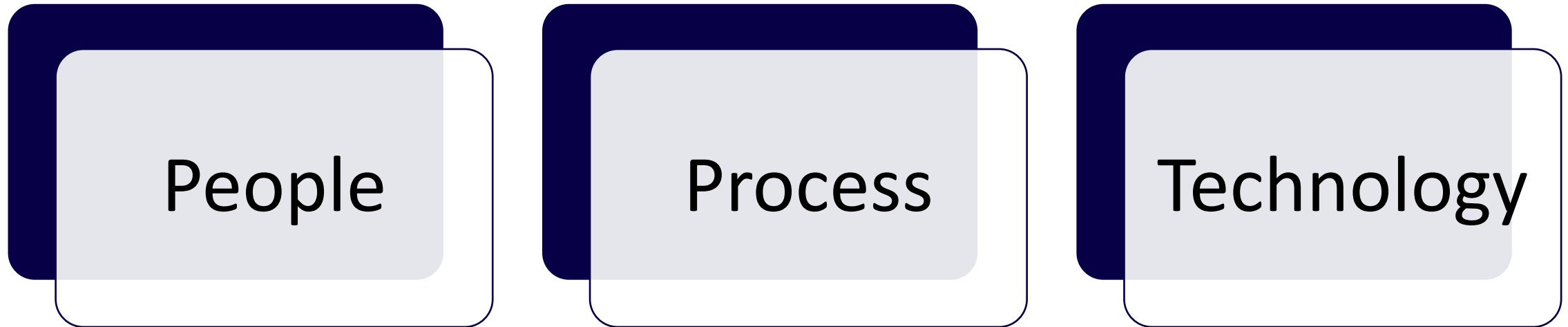


PROCESSES ARE BASED ON
GOOGLE FINOPS “EPICS”



EPICS ARE THE MOST
IMPORTANT ACTIVITIES

Google Cloud FinOps Fundamentals



Google's FinOps framework is build around the above criteria.

Google Cloud FinOps Fundamentals

Purpose of the Phase	Google Cloud Epics
Inform – Provide Transparent, accurate and actionable data	- Allocate (labels, tags, metadata) - Reporting KPI - Reporting Costs - Forecasting
Optimize – Create efficiency in the organization with controls	- Product Efficiency - Pricing Efficiency - Guardrails
Operate – Maintain the collaboration, constant innovation with incentivizing and measurements	- Onboard Workloads - Educate - Incentivize - Measure Value - Funding

Google Cloud FinOps Fundamentals

FinOps Whitepaper

The FinOps framework for Google Cloud Platform (GCP), covering the teams required, the processes and behaviors to embrace, the outcomes expected, and an approach to implement it.



Getting started with FinOps on GCP



Authors:
Sam Moss, Kinjal Tanna,
Tan-Minh Truong

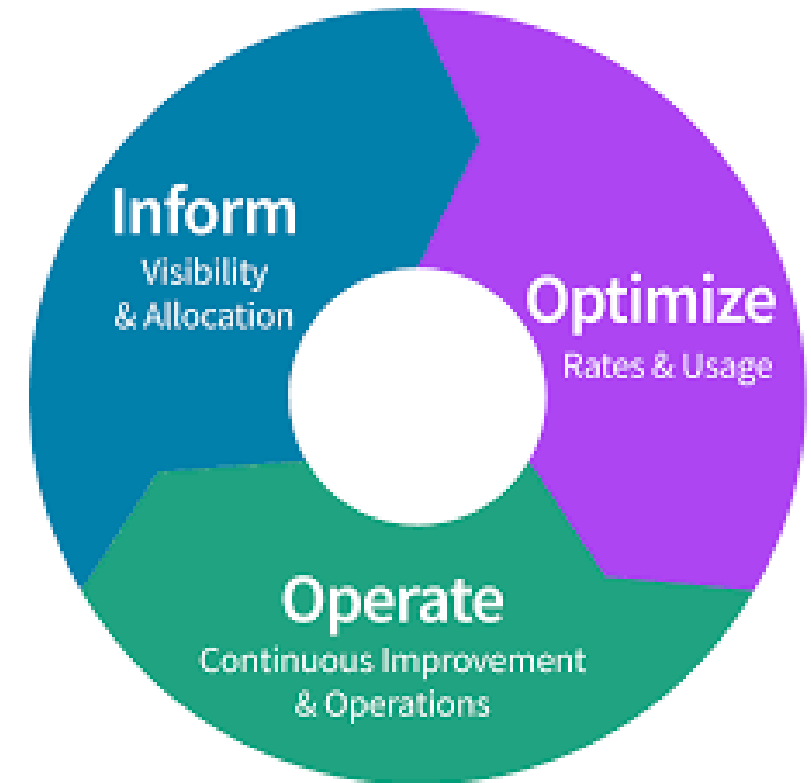
Google Cloud Cost Optimization

How to manage costs on Google Cloud

Google Cloud FinOps Fundamentals

Cost Optimization

- Cost Optimization is an exercise in which stakeholders have a continuous discipline to drive spending and cost reduction, while maximizing business value.
- FinOps plays a critical role in cost optimization with Cloud Spend.
- Understand the consumption model of cloud



Google Cloud FinOps Fundamentals

Building Blocks

Google Cloud Building Blocks of FinOps

- Accountability & Enablement
- Measurement & Realization
- Cost Optimization
- Planning & Forecasting
- Tools & Accelerators



Cost Optimization

Areas of common concern to for reducing costs.

- Remove/delete unused snapshots, disks for Compute Engine
- Delete external Ips that are not needed “gcloud compute addresses list”
- Check SQL instances and remove unneeded snapshots, instances, etc.
- Perform Quarterly rightsizing exercises. (Review bill)
- Maximize Committed Use Discounts and plan Sustained Use Discounts
- Consider On Demand or Ephemeral instance

Cost Optimization

- Setup Lifecycles in Cloud Storage to move data to appropriate class
- Visualize your monitoring with Cloud Operations Dashboards
- Define Tagging and Labeling standards to enable cost center chargeback
- Constantly measure with clear KPIs and metrics
- Kubernetes is challenging, consider the best practices to reduce cost challenges
- Train teams about FinOps and encourage a stakeholder responsibility

Cost Optimization

- Filter Network Packets to reduce bandwidth costs thru use of Cloud Logging, Network Service Tiers and by identifying your top talkers. (egress costs)
- Tune your BigQuery data queries. (Controls, partitioning, Flex Slots, streaming inserts)



Google Cloud FinOps Fundamentals

Service Options

Compute (IaaS/PaaS)	Storage and Databases	Networking/Content Distribution	Big Data and IOT
Compute Engine	Cloud Storage	Virtual Private Cloud (VPC)	BigQuery
Google Kubernetes Engine (GKE)	Cloud Bigtable	Cloud DNS	Cloud DataFlow
App Engine	Cloud SQL	Cloud NAT	Cloud DataProc
Cloud Functions	Cloud Spanner	Cloud VPN	Cloud DataLab
Cloud Run	Cloud Datastore	Load Balancing	Cloud Pub/Sub
	Persistent Disk	Cloud Interconnect	Google Data Studio
	FileStore	Cloud DNS	Genomics
	Firebase (Mobile)		Cloud IOT

Google Cloud FinOps Fundamentals

Service Options

ML/AI	Developer Tools	Monitoring and Mgmt.	Security and Access
Cloud ML Engine	Artifact Registry	Cloud Operations	Cloud IAM
Cloud Jobs API	Container Registry	Cloud APIS	Cloud Armor
Cloud Natural Language API	Cloud Source Repositories	Cloud Console	Cloud Key Management
Cloud Speech API	Cloud Tasks	Cloud Shell	Binary Authorization
Cloud Vision API	Cloud Code	Cost Management	Secrets Manager
Cloud Translation API	Cloud SDK	Billing Management	Cloud Data Loss Prevention API
Cloud Video Intel API	Apigee	Budgets	Security Command Center
TensorFlow	Anthos	Private Catalog	Chronicle
AI Infrastructure		Migration Options	

Google Cloud Resource Management

Infrastructure, Organizations

Google Cloud FinOps Fundamentals

Resource Management (Resource Manager)

- Resource Manager effectively provides a container for managing your resources in Google Cloud
- Hierarchal Organization (Organization, Projects, Folders and Resources)
- IAM Control
- Enforce Policies
- API management
- Cost Management

Cloud Infrastructure Selection

Segmentation of Cloud – Regions, Zones

Google Cloud FinOps Fundamentals

Google Cloud Platform resources are organized by Regions and Zones.

Regions – Collection of Zones

- Specific location to run resources
- Connected by Googles global and meshed backbone

Zones – Isolated deployment areas in a region.

- Resource can be zonal, regional or multi regional

(Pricing will vary for each service and international regions)

Google Cloud FinOps Fundamentals

Understanding Regions and Zones

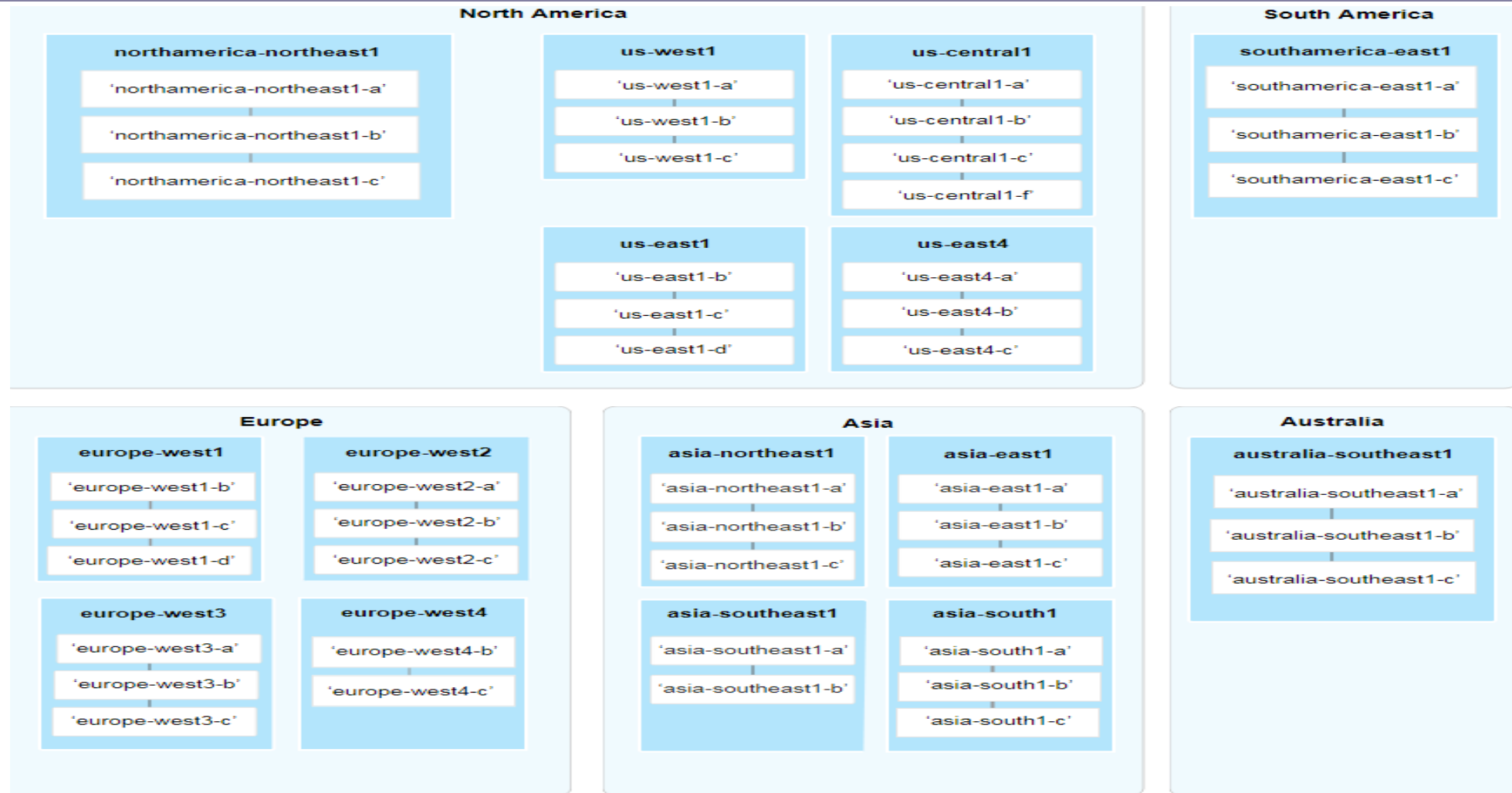
- Costing needs to be addressed by the FinOps Team
- Regions are independent geographic areas that consist of zones
- Locations within regions tend to have round-trip network latencies of under 5ms on the 95th percentile
- Zones are a deployment area for Cloud Platform resources within a region
- Zones should be considered a single failure domain within a region
- Some Cloud Platform services are located at a multi-regional level rather than the more granular regional or zonal levels

Choose your region and zones based on the following.

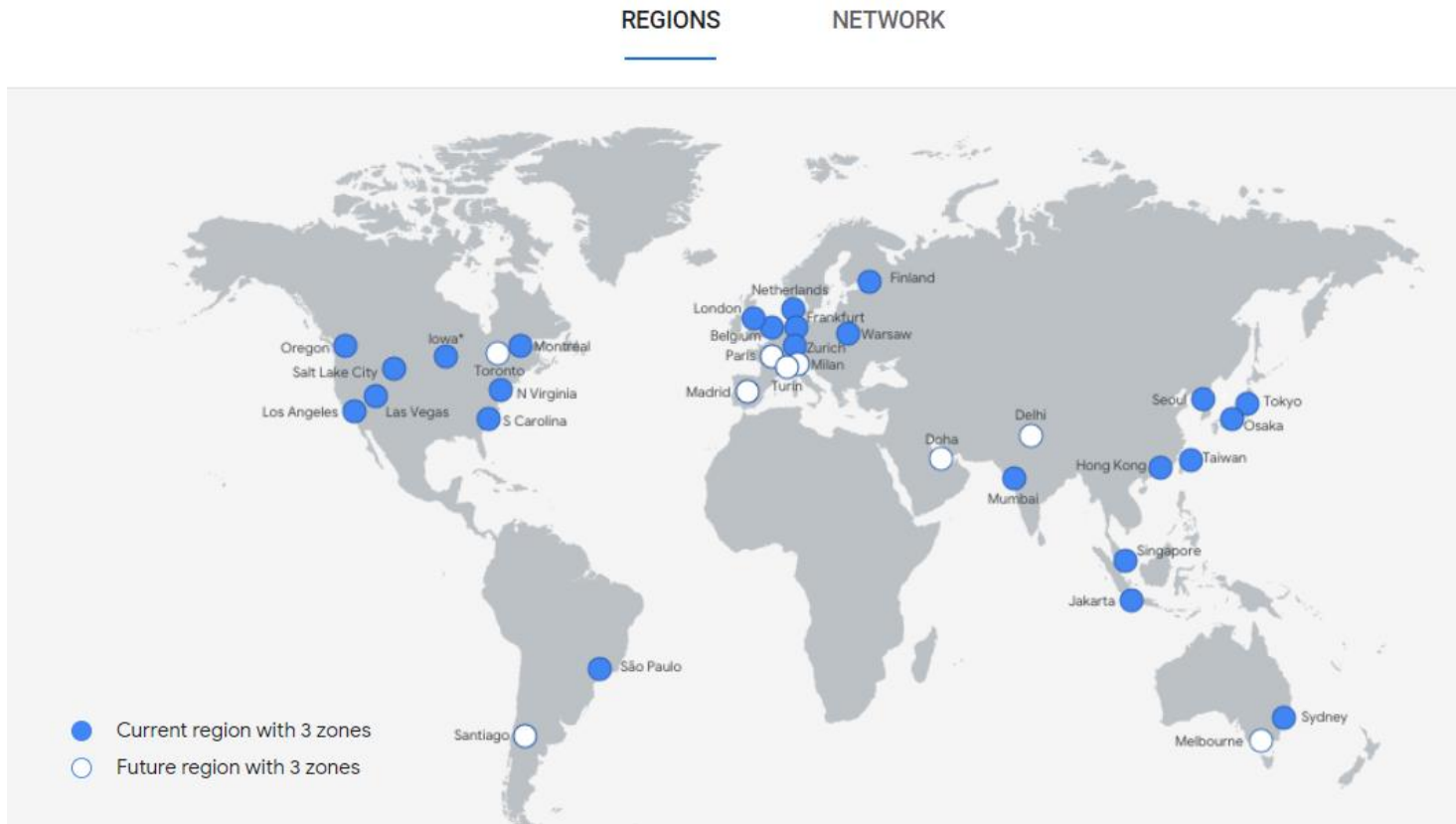
- Compliance Requirements
- Costing
- Services Available
- Application Latency
- Customer Location

Google Cloud FinOps Fundamentals

Regions and Zones



Google Cloud FinOps Fundamentals



25
REGIONS

76
ZONES

144
NETWORK EDGE LOCATIONS

AVAILABLE IN
200+
COUNTRIES AND TERRITORIES

COMING SOON! Google Cloud will continue expanding into the following regions: Doha (Qatar), Toronto (Canada), Melbourne (Australia), Delhi (India), Paris (France), Milan (Italy), Santiago (Chile), Madrid (Spain) and Turin (Italy).

<https://cloud.google.com/about/locations>

GCP Cloud Hierarchy

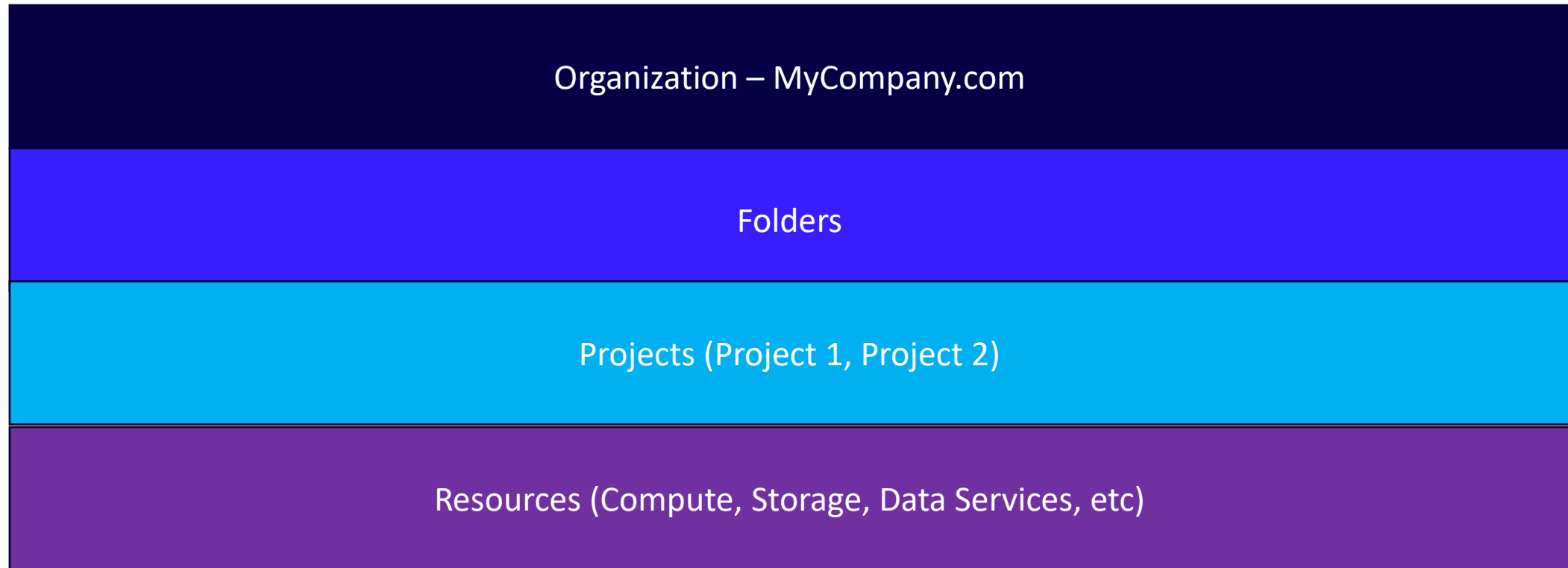
Organizations, Projects and Folders

Google Cloud FinOps Fundamentals

The hierarchy of GCP is broken into the following structure:

- Organizations
- Folders
- Projects
- Resources

Google Cloud FinOps Fundamentals



Google Cloud FinOps Fundamentals

Organizations

- The Organization resource is the root node of the Google Cloud resource hierarchy and all resources that belong to an organization are grouped under the organization node.
- The Organization resource is associated with a Google Workspace or Cloud Identity account.
- The Google Workspace super admin is the individual responsible for domain ownership verification and the contact in cases of recovery.

Google Cloud FinOps Fundamentals

Google Cloud resources are organized hierarchically.

- Organization is the top of the hierarchy and does not have a parent.
- Projects are the first level, and they contain other resources.
- From a FinOps Perspective we assign(link) a “billing account”
- The Cloud Billing account is also linked to a Google payments profile, which stores information like name, address, and payment methods.

Google Cloud FinOps Fundamentals



AN ORGANIZATION RESOURCE IS AVAILABLE FOR WORKSPACE AND CLOUD IDENTITY CUSTOMERS.



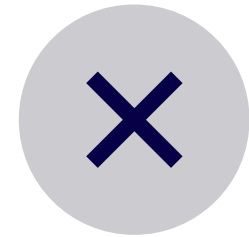
LINK YOUR ORG DOMAIN TO GCP.



THINK OF AN ORGANIZATION AS A HIERARCHY.



SET ACCESS CONTROL AND CONFIGURATION SETTINGS AT THE ORGANIZATION OR PROJECT LEVEL



BILLING ACCOUNTS, PROJECTS, AND RESOURCES ARE NOT DELETED WHEN AN EMPLOYEE LEAVES THE COMPANY. FOLLOWS CORPORATE LIFECYCLE.

Projects



Google Cloud FinOps Fundamentals

Projects

A Project facilitates organization of services and objects and uses this method of segmentation for billing and accounting.



Project info

Project name

My Python Hello World

Project ID

my-python-hello-world-191118

Project number

452326268329



[Go to project settings](#)

Google Cloud FinOps Fundamentals

Projects in GCP

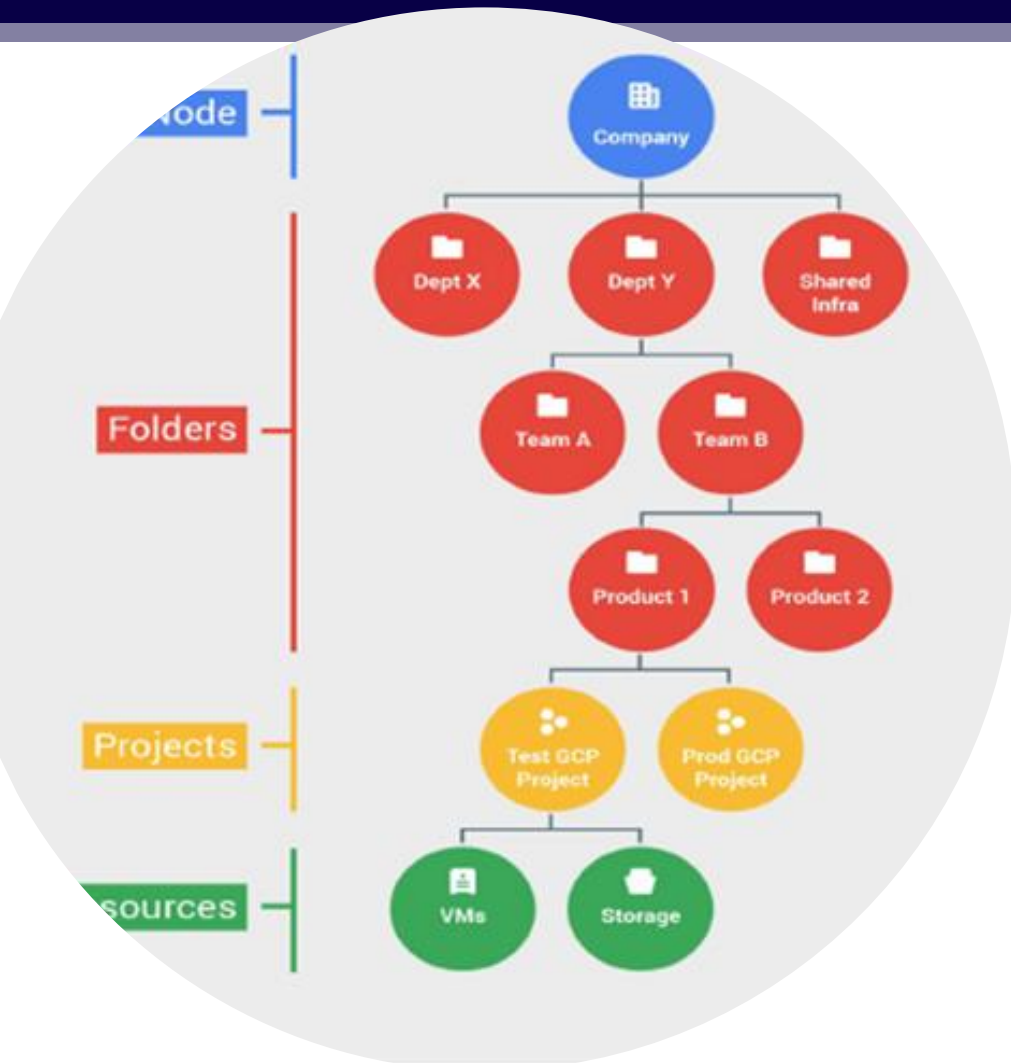
Projects

All Google Cloud Platform services are associated with a project that is used to:

- Track resource and quota usage
- Enable billing
- Manage permissions and credentials
- Enable services and APIs

```
jholbrookln@projectidgcpcourse1:~$ gcloud projects list
PROJECT_ID          NAME                PROJECT_NUMBER
gcpcourse1234        gcpcourse1234       824698542660
my-awesome-project-198000 My Awesome Project  577767719453
my-test-project-2-192418 My Test Project 2   174359988401
projectidgcpcourse1  Project AWS and GCP Course 409356428059
projectnetworking101 projectnetworking101 251983216344
test123-203018       test123             918409078391
jholbrookln@projectidgcpcourse1:~$
```


Google Cloud FinOps Fundamentals



Projects and Hierarchy

- Folders are also introduced when you use Cloud IAM.
- The Cloud IAM Folders feature lets you assign policies to resources at a level of granularity you choose.
- The resources in a folder can share IAM policies
- **Google Cloud IAM is comparable to AWS Directory Service.**

Demo - Google Cloud Resource Management

Demo – Resource Management

Google Cloud FinOps Fundamentals



Whiteboard – Organizing Resources

Whiteboard - Lets cover FinOps related concerns

Google Cloud FinOps Fundamentals

FinOps Management with GCP



Whiteboard – FinOps Management with GCP

Whiteboard - Lets cover FinOps management

Demo – Resource Organization

Demo

Google Cloud FinOps Fundamentals



Demo – Billing

Demo

Google Cloud FinOps Fundamentals



Demo – Invoices and Reporting

Demo

Google Cloud FinOps Fundamentals



Demo – Pricing

Demo

Google Cloud FinOps Fundamentals



Tag and Label Management

Tags, Labels and Metadata

Google Cloud FinOps Fundamentals

Metadata

Compute Engine defines a set of default metadata entries that provide information about your virtual machine (VM) instance or project.

Default metadata is always defined and set by the server.

Custom metadata

Key	Value
finops	demo1

[EQUIVALENT REST](#)

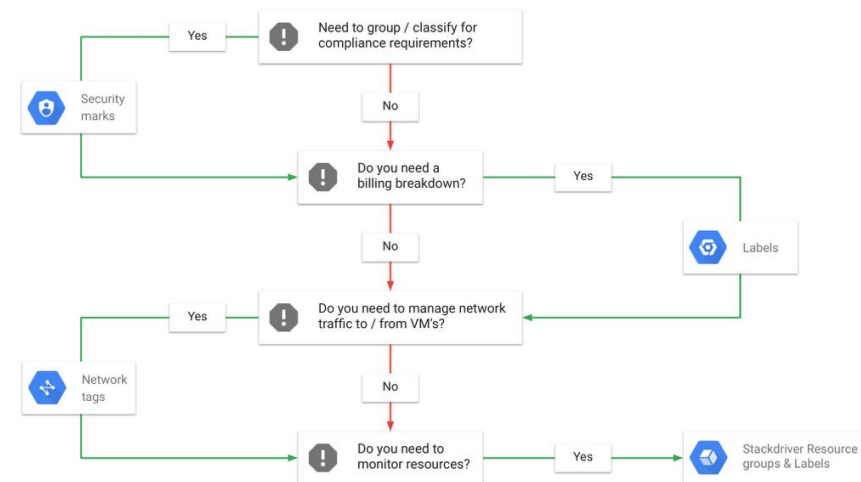
Google Cloud FinOps Fundamentals

Tags or Labels

GCP has been changing its reference to tags and labels.

GCP now is using tags to refer to network tags (or target tags as firewall rules call them) and labels to apply to labels (billing).

Once both were referenced as tags.



<https://cloud.google.com/blog/products/gcp/labelling-and-grouping-your-google-cloud-platform-resources>

Google Cloud FinOps Fundamentals

Labels

A label is a key-value pair that helps you organize your Google Cloud resources.

You can attach a label to each resource, then filter the resources based on their labels.

Information about labels is forwarded to the billing system, so you can break down your billed charges by label.

Only certain services support labels.

The screenshot shows the 'Create an instance' page in the Google Cloud console. On the left, there are three options: 'New VM instance' (selected), 'New VM instance from template', and 'New VM instance from machine image'. The 'New VM instance' option is expanded, showing sub-options: 'Create a single VM instance from scratch', 'Create a single VM instance from an existing template', and 'Create a single VM instance from an existing machine image'. The 'Marketplace' option is also visible. The main configuration area includes: 'Name' (Instance-1), 'Labels' (+ ADD LABELS), 'Region' (us-central1 (Iowa)), 'Zone' (us-central1-a), 'Machine configuration' (Machine family: GENERAL PURPOSE, Machine type: e2-medium (2 vCPU, 4 GB memory)), 'Display device' (checkbox: Enable display device), 'Confidential VM service' (checkbox: Enable the Confidential Computing service on this VM instance), and 'Container' (checkbox: Enable containerd on this VM instance). A 'Monthly estimate' of \$25.46 is shown on the right. At the bottom right, there are 'SAVE' and 'CANCEL' buttons.

<https://cloud.google.com/resource-manager/docs/creating-managing-labels>

Google Cloud FinOps Fundamentals

Tags vs Labels

Tags are a way to group resources together to have a *network policy* applied to them. (Organization, folder or project)

When a tag key-value pair is attached to a resource, it will be inherited by all the resource's descendants.

A label is used to groups resources together for *billing and searching purposes*

Manage labels



Labels entered here will be assigned at the time of instance creation. Labels for an existing instance may be edited on the instance details page.

Key 1
finops

Value 1
team1



+ ADD LABEL

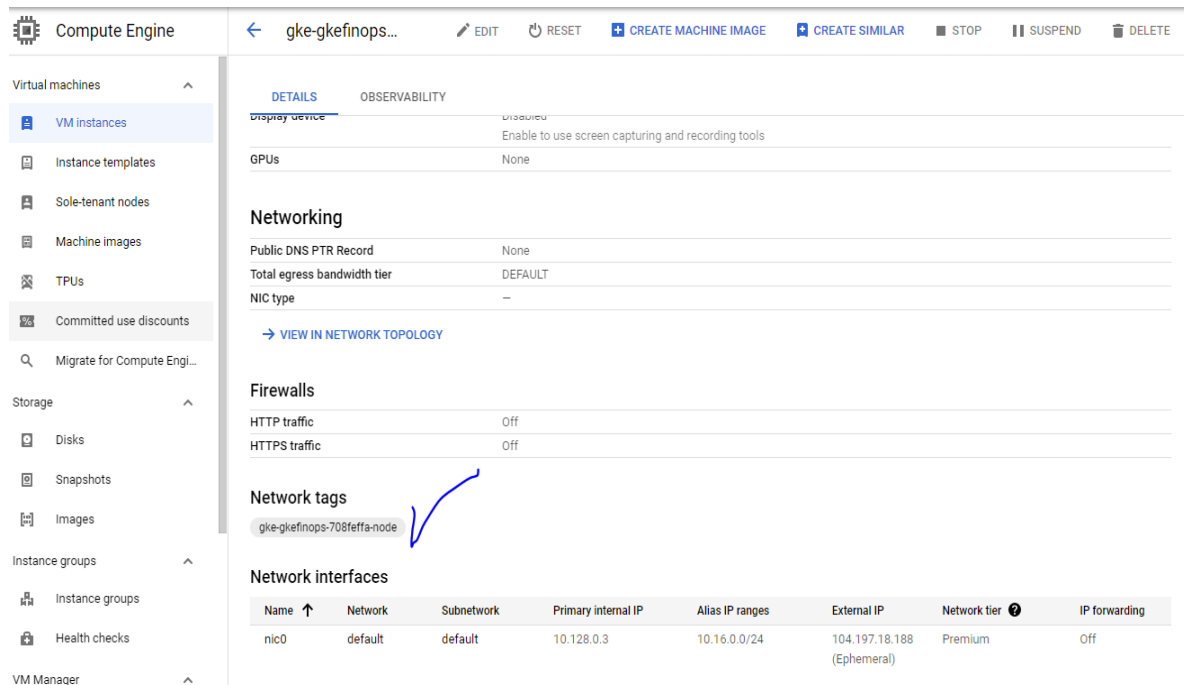
Google Cloud FinOps Fundamentals

Tags vs Labels

Tags	Labels
Tags are a way to group resources together to have a network policy applied to them. (Organization, folder or project)	A label is used to groups resources together for billing and searching purposes
Primary Purpose is to group resources for network tagging	Primary purpose is track resource usage
To Configure go to IAM	To Configure go to resource and select for billing/search

Google Cloud FinOps Fundamentals

Tags vs Labels



The screenshot shows the Google Cloud Console interface for a Compute Engine instance named 'gke-gkefinops...'. The left sidebar lists various services, and the main panel displays the instance details. The 'Network tags' section is highlighted with a blue checkmark, indicating that tags are used for instance identification.

Virtual machines

- VM instances
- Instance templates
- Sole-tenant nodes
- Machine images
- TPUs
- Committed use discounts
- Migrate for Compute Engi...

Storage

- Disks
- Snapshots
- Images

Instance groups

- Instance groups
- Health checks

VM Manager

DETAILS | OBSERVABILITY

Display device | Disabled
Enable to use screen capturing and recording tools

GPUs | None

Networking

Public DNS PTR Record | None

Total egress bandwidth tier | DEFAULT

NIC type | --

[VIEW IN NETWORK TOPOLOGY](#)

Firewalls

HTTP traffic | Off

HTTPS traffic | Off

Network tags

gke-gkefinops-708feffa-node

Network interfaces

Name	Network	Subnetwork	Primary internal IP	Alias IP ranges	External IP	Network tier	IP forwarding
nic0	default	default	10.128.0.3	10.16.0.0/24	104.197.18.188 (Ephemeral)	Premium	Off

Manage labels

Labels entered here will be assigned at the time of instance creation. Labels for an existing instance may be edited on the instance details page.

Key 1 *
development

Value 1
ecommerce

+ ADD LABEL

Google Cloud FinOps Fundamentals

Label Examples

Application Name

Region Name

Business Unit Name

Resource Creator

Billing or BU Codes

Resource Owner

Classification

Labels ?

development : application1

legal : compliance

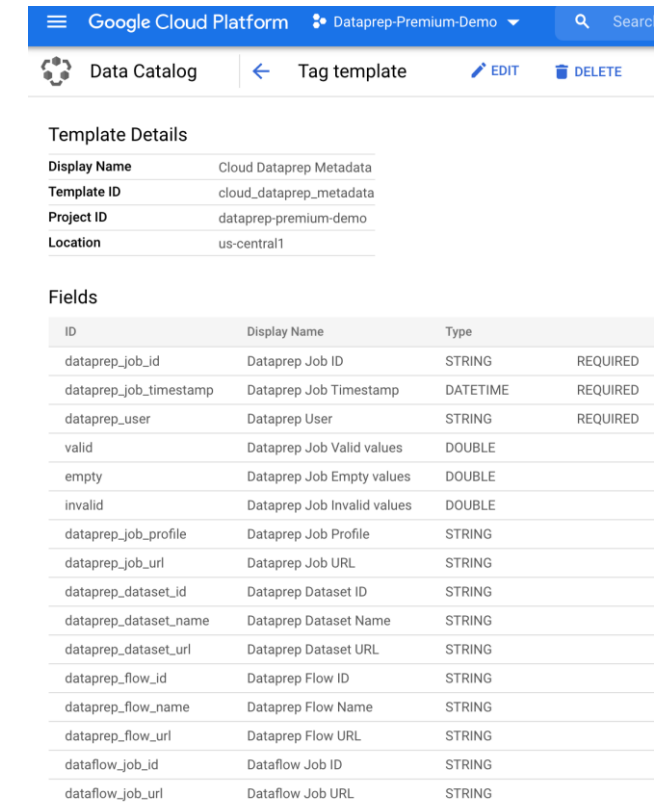
+ ADD LABELS

Data Catalog

- A Data Catalog is responsible for the maintenance of data assets. It records discovery, organization, and dataset description.
- The Data Catalog enables the data analysts, scientists, and other consumers to query and use data from the datasets and understand.
- A data catalog is responsible for maintaining an inventory of data assets through the discovery, description, and organization of datasets.
- It provides a meaningful context to enable data analysts, scientists, and other data consumers to search and to be able to understand a relevant dataset to extract business value.

Data Catalog

- Data Catalog is managed by Google Cloud and is a centralized service.
- The data catalog builds and manages an optimized index for searching data assets such as datasets, views, tables, files, streams, and spreadsheets.
- The metadata of these assets to build up the index.
- Update or storage of assets causes the metadata to be updated or created and later be changed in the source systems.



The screenshot displays the Google Cloud Platform interface for the Data Catalog. The top navigation bar shows 'Google Cloud Platform' and 'Dataprep-Premium-Demo'. Below this, the 'Data Catalog' section is active, showing a 'Tag template' for 'Cloud Dataprep Metadata'. The 'Template Details' section lists the following information:

Display Name	Cloud Dataprep Metadata
Template ID	cloud_dataprep_metadata
Project ID	dataprep-premium-demo
Location	us-central1

Below the details, the 'Fields' section contains a table with the following data:

ID	Display Name	Type	Required
dataprep_job_id	Dataprep Job ID	STRING	REQUIRED
dataprep_job_timestamp	Dataprep Job Timestamp	DATETIME	REQUIRED
dataprep_user	Dataprep User	STRING	REQUIRED
valid	Dataprep Job Valid values	DOUBLE	
empty	Dataprep Job Empty values	DOUBLE	
invalid	Dataprep Job Invalid values	DOUBLE	
dataprep_job_profile	Dataprep Job Profile	STRING	
dataprep_job_url	Dataprep Job URL	STRING	
dataprep_dataset_id	Dataprep Dataset ID	STRING	
dataprep_dataset_name	Dataprep Dataset Name	STRING	
dataprep_dataset_url	Dataprep Dataset URL	STRING	
dataprep_flow_id	Dataprep Flow ID	STRING	
dataprep_flow_name	Dataprep Flow Name	STRING	
dataprep_flow_url	Dataprep Flow URL	STRING	
dataflow_job_id	Dataflow Job ID	STRING	
dataflow_job_url	Dataflow Job URL	STRING	

Best Practices

- Have an enterprise standard for labels and maintain a policy for labels
- *Search using labels*
- *BigQuery supports labels, so use them*
- *Label your GKE Nodes, Compute Engine Nodes, Cloud Storage Buckets, SQL Instances, etc*

Labels



Labels can help you organize your Cloud SQL instances. [Learn more](#)

production: ecommerce

(Not saved) ▼

[ADD LABEL](#)

Demo – Label Management

Demo

Google Cloud FinOps Fundamentals



Demo – Budgets

Demo

Google Cloud FinOps Fundamentals



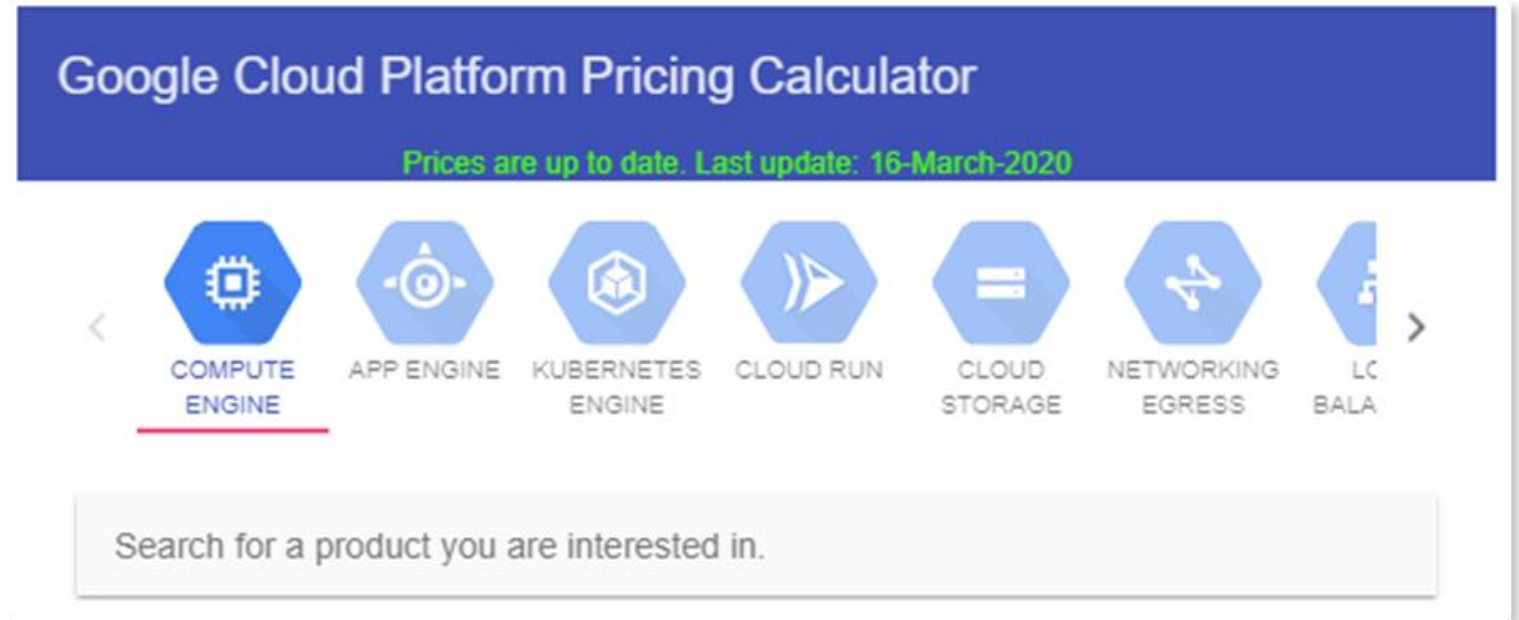
Pricing Calculator

Understanding how to estimate costs of cloud resources

Google Cloud FinOps Fundamentals

Pricing/Cost Estimating

You can easily
estimate the cost
of GCP Services by
going to the
Pricing Calculator



Google Cloud FinOps Fundamentals

Pricing Calculator

Google Cloud Platform Pricing Calculator

Prices are up to date. Last update: 11



Estimate ¹

BigQuery

ON-DEMAND FLAT-RATE

Table Name

Name

?

Storage Pricing

Storage

GB

▼

?

Streaming Inserts

MB

▼

?

Query Pricing

Queries

TB

▼

?

Google Cloud FinOps Fundamentals

Pricing/Cost Estimating

Select your product and estimate will appear for usage.

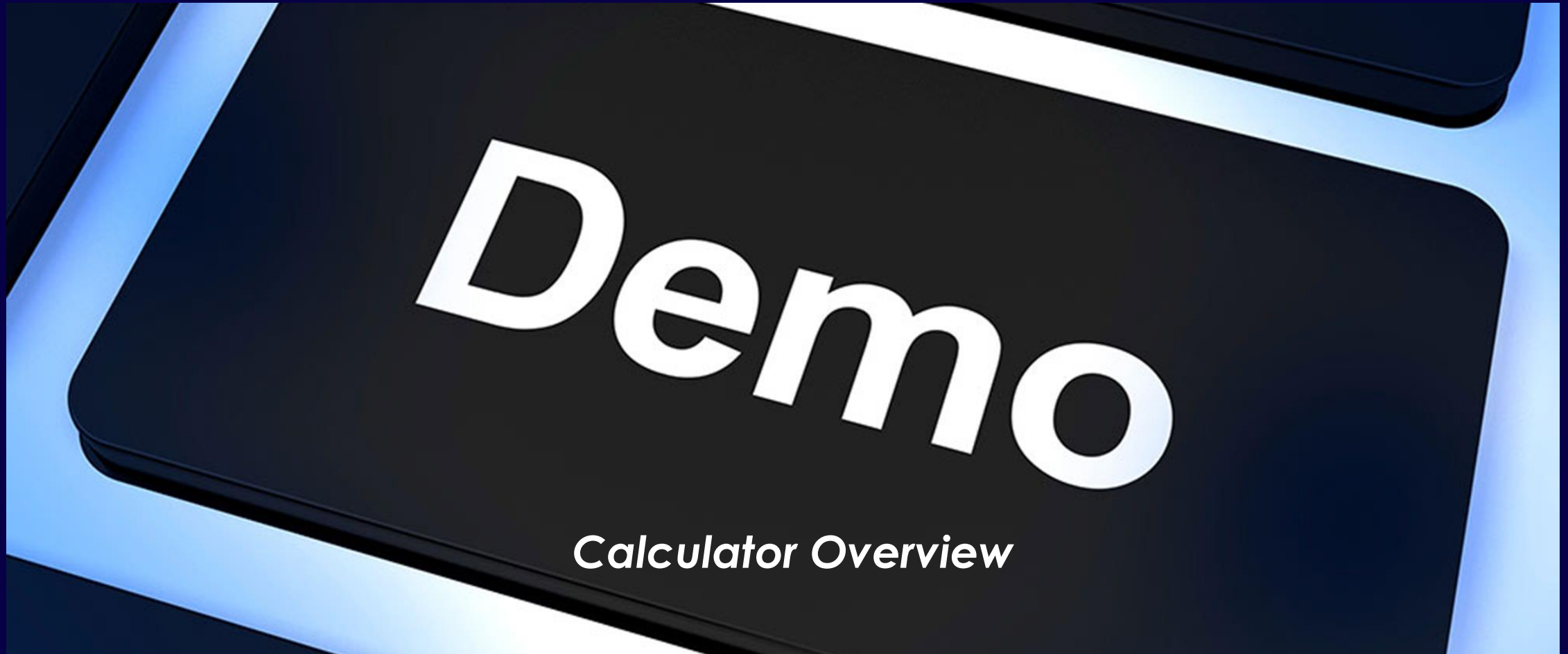
Variables differ from each service.

Pricing Calculator is here. <https://cloud.google.com/products/calculator/>

Pricing Calculator

Demo

Google Cloud FinOps Fundamentals



Whiteboard – Putting it all together

Main Section Topics

Google Cloud FinOps Fundamentals

Putting it all Together



Hands-on exercise

A photograph of two women in an office environment. One woman, with blonde hair and glasses, is pointing at a computer monitor displaying code. The other woman, with dark curly hair, is looking at the screen. The background shows a bright office with windows and a potted plant.

Estimate Cloud Costs with Pricing Calculator

Section Summary

Section 2 : Google Cloud Tools/Services FinOps



2

Section

Section Summary

- Google's FinOps framework is build around People, Process and Technology. (Epics)
- Google Cloud Building Blocks of FinOps are:
 - 1. Accountability & Enablement
 - 2. Measurement & Realization
 - 3. Cost Optimization
 - 4. Planning & Forecasting
 - 5. Tools & Accelerators
- Google Clouds infrastructure is broken into Regions and Zones which can have different services and costing associated with them.
- The hierarchy of GCP is broken into the following structure: Organizations, Folders, Projects and Resources which can provide an efficient way to isolate a business units cloud spending.
- GCP now is using tags to refer to network tags (or target tags as firewall rules call them) and labels to apply to labels (billing).
- The Cloud Billing account is linked to a Google payments profile, which stores information like name, address, and payment methods.

Section Review Questions

Section 2 : Google Cloud Tools/Services FinOps



Review Questions

Which of following answers would not be a building block of GCP FinOps (Select One)

- Accountability & Enablement
- Measurement & Realization
- Cost Optimization
- Planning & Forecasting
- Itemization and Cataloging

Review Questions

Which of following answers would not be a building block of GCP FinOps (Select One)

- Accountability & Enablement
- Measurement & Realization
- Cost Optimization
- Planning & Forecasting
- Itemization and Cataloging

Review Questions

A _____ is a physical location around the world where GCP clusters their data centers. (Select One)

- Region
- Edge
- Availability Zone
- Local Zone

Review Questions

A _____ is a physical location around the world where GCP clusters their data centers. (Select One)

- Region
- Edge
- Availability Zone
- Local Zone

Review Questions

Google Cloud has a strict well-defined hierarchy. Which of the following would be used with Google Workspace for a top-down lifecycle? (Select One)

- Projects
- Organization
- Folders
- Resources

Review Questions

Google Cloud has a strict well-defined hierarchy. Which of the following would be used with Google Workspace for a top-down lifecycle? (Select One)

- Projects
- **Organization**
- Folders
- Resources

Review Questions

Which of the following three tools would be used with Google Cloud FinOps Tasks?
(Select Three)

- Billing Exports (BigQuery)
- Cloud Hierarchy
- Labels and Tags
- Vertex/AI
- Deployment Manager

Review Questions

Which of the following three tools would be used with Google Cloud FinOps Tasks?
(Select Three)

- Billing Exports (BigQuery)
- Cloud Hierarchy
- Labels and Tags
- Vertex/AI
- Deployment Manager

Review Questions

Which of the following answers would be used with Google Cloud FinOps related best practices? (Select Three)

- Deploy resources with 3rd party tools to maintain costs
- Define Tagging and Labeling standards to enable cost center chargeback
- Constantly measure with clear KPIs and metrics by implementing network tags
- Kubernetes is challenging, consider the best practices to reduce cost challenges
- Train your teams about FinOps and encourage a stakeholder responsibility

Review Questions

Which of the following answers would be true with Google Cloud FinOps related best practices? (Select Four)

- Deploy resources with 3rd party tools to maintain costs
- Define Tagging and Labeling standards to enable cost center chargeback
- Constantly measure with clear KPIs and metrics by implementing network tags
- Kubernetes is challenging, consider the best practices to reduce cost challenges
- Train your teams about FinOps and encourage a stakeholder responsibility

Review Questions

BigQuery is a data warehouse that has many features to help maintain costs. Which of the following would be ways to implement cost controls. (Select Three)

- Partitioning
- Flex Slots
- Streaming inserts
- API
- Turn off Audit Logging

Review Questions

BigQuery is a data warehouse that has many features to help maintain costs. Which of the following would be ways to implement cost controls. (Select Three)

- Partitioning
- Flex Slots
- Streaming inserts
- API
- Turn off Audit Logging

Section 3

Managing Service Costs on GCP

Cost Management

Section Overview

OPTIONS FOR MANAGING COST ON GOOGLE	WHITEBOARD - CASE STUDY	FORECASTING CLOUD SPEND	COMPUTE SERVICES DISCOUNT MODELS	DEMO - ACTIVE ASSIST	DEMO - INSTANCE DISCOUNTS	STORAGE SERVICES COST MANAGEMENT
DEMO - FINDING STORAGE WASTE	DEMO - UNDERSTANDING KUBERNETES COSTS	COST OPTIMIZATION CHALLENGES WITH DEVOPS ENVIRONMENTS	BIGQUERY PRICING MODEL	ACCOUNT BILLING MANAGEMENT	DEMO - BILL EXPORT AND BIGQUERY	VISUALIZING YOUR DATA
DEMO - DATA STUDIO	CLOUD MANAGEMENT - CLOUD OPERATIONS	DEMO - CREATE AN OPERATIONS DASHBOARD	DEMO - BUDGET COST CONTROLS	DEMO - BUDGET NOTIFICATIONS	DEMO - QUOTAS	CLOUD LANGUAGE
	WHITEBOARD - CONSIDERATIONS AND SUMMARY	SECTION REVIEW	SECTION REVIEW QUESTIONS	RESOURCES	THANK YOU	

Options for Managing Cost on GCP

Understanding the options for services.

Google Cloud FinOps Fundamentals

Manage Costs – Tools, Services, Best Practices

- Active Assist – Recommendations
- Set Up Budgets and Alerts
- Setup Hierarchy Correctly
- Validate Chargeback Structure
- Sized VM Instances with correct discount models
- Select Appropriate Regions
- Billing details for each service

☒ Enable Cloud Monitoring  ✓

System 

☐ Enable Managed Service for Prometheus  

Service mesh

☐ Enable Anthos Service Mesh 

Other

☐ Enable Cloud TPU 

☐ Enable Kubernetes alpha features in this cluster 

☒ Enable GKE usage metering  ✓

BigQuery dataset *
FinOpsDemo  

☐ Enable network egress metering 

☒ Enable resource consumption metering 

Google Cloud FinOps Fundamentals

Manage Costs – Tools, Services, Best Practices

- Setup custom data-driven dashboards (Operations, 3rd party)
- Set your KPIs and Identify Metrics
- Follow Kubernetes Best Practices
- Follow BigQuery Best Practices
- Use 3rd Party if needed
- Labelling your resources
- Ensure Visibility into Costs
- Setup IT Cost Assessment
- Train your teams and stakeholders

Google Cloud FinOps Fundamentals

3rd party Tools FinOps and Cost Management

Third-party cost management tools can address the functional limitations of first party tools. They can also provide multi-cloud cost management. (May also see an ROI)

- CastAI
- AppTio
- Harness
- Spot (NetApp)

Forecasting Cloud Spend

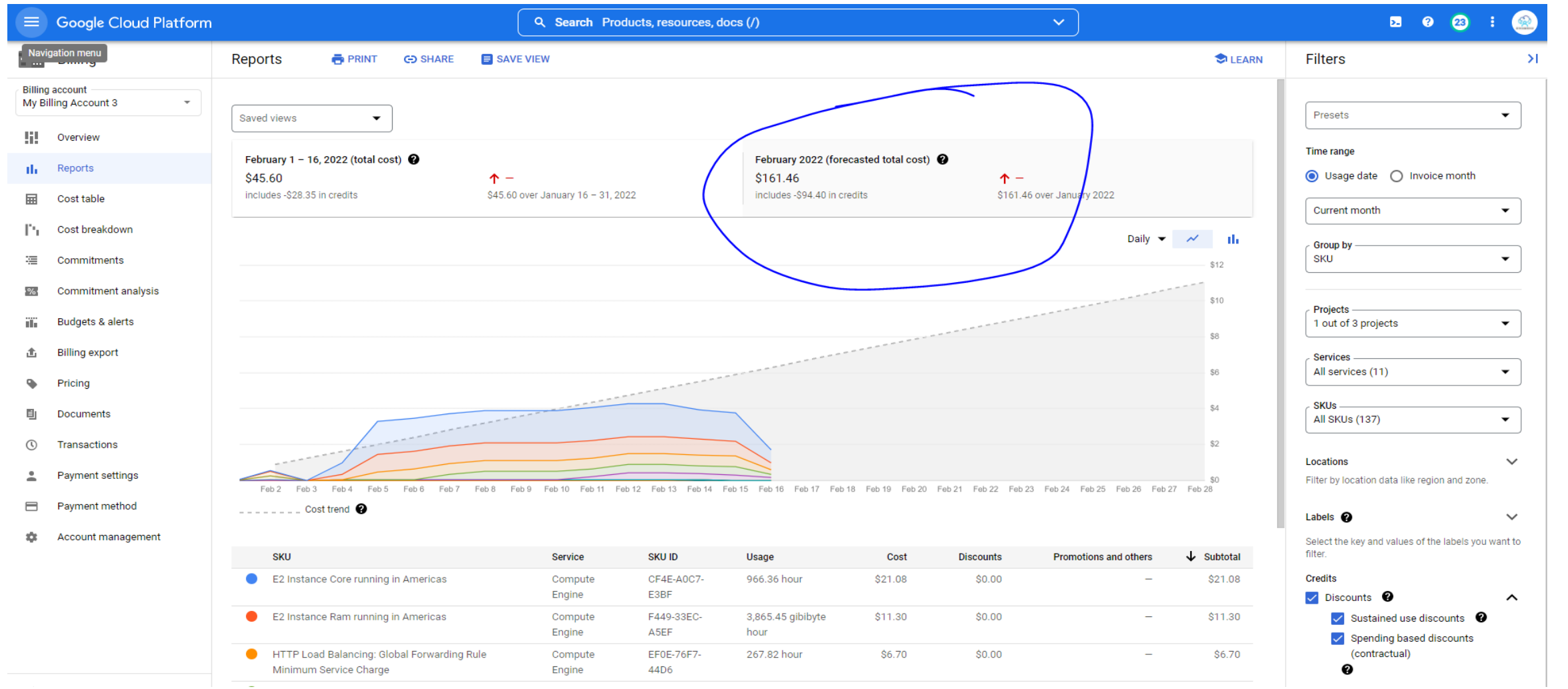
Best Practices on GCP

Google Cloud FinOps Fundamentals

Forecasting

- View Billing (Forecasted)
- Setup Budgets (thresholds)
- Setup Alerts
- Audit Logs (Export and Activity logs are deprecated)
- Monitor Resources (Cloud Operation)

Google Cloud FinOps Fundamentals



Google Cloud FinOps Fundamentals

Billing

Billing account
My Billing Account 3

Overview

Reports

Cost table

Cost breakdown

Commitments

Commitment analysis

Budgets & alerts

Billing export

Pricing

Documents

Transactions

Payment settings

Payment method

Account management

Budgets & alerts

CREATE BUDGET

DELETE

LEARN

Budgets track expenses within a Google Cloud Platform project or billing account. Your budget can be a specified amount or based on previous spend. You can set alerts to notify billing admins and users when a budget goes over a specified amount.

Setting a budget does not cap resource or API consumption. [Learn more.](#)

Filter

Enter property name or value

?

☐	Budget name ↑	Budget period	Budget type	Applies to	Trigger alerts at	Spend and budget amount
☐	FinOps Notification	Monthly	Specified amount	This billing account	50%, 90%, and 100%	\$55.72 / \$1,000.00 Includes -\$28.41 credit
☐	Monthly Alerts	Monthly	Specified amount	This billing account	50%, 90%, and 100%	\$55.72 / \$100.00 Includes -\$28.41 credit

Google Cloud FinOps Fundamentals

Forecasting

Daily usage reports provide usage information about the following resources:

1. Virtual machines
2. Persistent disks
3. Images
4. Snapshots
5. Static IP addresses
6. Load balancers

Google Cloud FinOps Fundamentals

 **Compute Engine**

Virtual machines ^

 VM instances

 Instance templates

 Sole-tenant nodes

 Machine images

 TPU

 Committed use discounts

 Migrate for Compute Engine...

Storage ^

 Disks

 Snapshots

Settings

Daily usage reports export as CSV files to a Cloud Storage location of your choice. A Google Service Account will be granted write access to this location. [Learn more](#)

☒ Enable usage export

Cloud Storage bucket * ? [BROWSE](#)

Report prefix ?

Default location

Newly created resources, like VM instances, will be deployed to the selected region and/or zone

Region ? Zone ?

Whiteboard – Case Study

Whiteboarding a scenario

Google Cloud FinOps Fundamentals

Case Study

- Large Fortune 1000 company
 - Migrated to GCP last year
 - Cloud Costs increasing 26% YoY
 - Uses a combination of Committed and On Demand Compute
 - GKE and Cloud SQL
 - Cloud Storage
-
- Customer has requested to identify areas to lower cloud costs with actions to remediate immediately.

Google Cloud FinOps Fundamentals

Putting it all Together



Compute Billing and Discount Models

Understanding Billing and Discounts

Google Cloud FinOps Fundamentals

Google Cloud offers a wide range of Compute Engine machine types. These include:

- General-purpose N1, N2, N2D and E2 machines that offer the best price-performance ratio.
- Memory-optimized VMs that offer higher memory per core, up to 12 TB.
- Compute-optimized machines that offer the highest performance per core for compute-intensive workloads.
- Shared-core machines for N1 and E2 VMs for a cost-effective way to run small, non-resource-intensive applications.

Google Cloud FinOps Fundamentals

Machine configuration

Machine family

- GENERAL-PURPOSE
- COMPUTE-OPTIMIZED
- MEMORY-OPTIMIZED
- GPU

Machine types for common workloads, optimized for cost and flexibility

Series

E2

CPU platform selection based on availability

Machine type

e2-medium (2 vCPU, 4 GB memory)



vCPU

1 shared core

Memory

4 GB

CPU platform

Automatic

GPUs

The number of attached GPUs affects the VM's maximum number of memory and CPUs.

[Learn More](#)

+ ADD GPU

^ CPU PLATFORM AND GPU

Monthly estimate

\$122.94

That's about \$0.17 hourly

Pay for what you use: No upfront costs and per second billing

Item	Monthly estimate
4 vCPU + 16 GB memory	\$152.43
10 GB balanced persistent disk	\$1.00
Sustained use discount	-\$30.49
Total	\$122.94

[Compute Engine pricing](#)

^ LESS

Google Cloud FinOps Fundamentals

Billing Model

Per-second billing, sustained use discounts

- 1 minute minimum and 1 sec increments

Preemptible instances

- Live at most 24 hours
- Can be pre-empted with a 30 second notification via API
- Discounted significantly

Google Cloud FinOps Fundamentals

Billing Discounts

VM vCPU and memory usage for each of these categories can receive one of the following discounts:

- Sustained use discounts
- Committed use discounts
- Discounts for preemptible VM instances

Iowa (us-central1) Monthly ☒ Hourly				
Item	On-demand price	Preemptible price	1 year commitment price	3 year commitment price
Predefined vCPUs	\$0.031611 / vCPU hour	\$0.006655 / vCPU hour	\$0.019915 / vCPU hour	\$0.014225 / vCPU hour
Predefined Memory	\$0.004237 / GB hour	\$0.000892 / GB hour	\$0.002669 / GB hour	\$0.001907 / GB hour

<https://cloud.google.com/compute/pricing>

Google Cloud FinOps Fundamentals

Discount Model – Sustained Use

- **Sustained use discounts** are applied on incremental use after you reach certain usage thresholds.
- This means that you pay only for the number of minutes that you use an instance.
- GCP automatically gives you the best price.

Google Cloud FinOps Fundamentals

Discount Model – Sustained Use

Sustained use discounts apply to the following resources:

- Predefined machine types
- The vCPUs and system memory for custom machine types
- The vCPUs and system memory occupied by sole-tenant nodes
- The 10% premium cost for sole-tenant nodes, even if the vCPUs and system memory in those nodes are covered by committed use discounts
- GPU devices

Google Cloud FinOps Fundamentals

Discount Model – Sustained Use

How does Sustained Use Discounts work? Incrementally.

Usage %	Charged	N1-standard Instance
0%-25%	100% of base rate	\$0.0475
25%-50%	80% of base rate	\$0.0381
50%-75%	60% of base rate	\$0.0286
75% -100%	40% of base rate	\$0.0190

Sustained use discounts are a separate line item combining all discounts of all machine types on a per project basis.

Google Cloud FinOps Fundamentals

Discount Model – Committed Use

- **Committed use discounts** are very useful for workloads with predictable resource needs.
- When you purchase a committed use contract, you purchase compute resources at a discounted price. Commitments are allocated over the course of each hour in a month.
- Committed use discounts apply to both vCPUs and memory.
- Commitment for 1 year or 3 years

Google Cloud FinOps Fundamentals

Discount Model – Committed Use

Commitments are assigned on a vCPU/RAM basis

Allocation of commitments is handled when your bill is generated, and your discount is applied in a very specific order:

1. To custom machine types
2. Sole-tenant node groups
3. Predefined machine types

Google Cloud FinOps Fundamentals

Discount Model – Capacity Reservations

- **Reserved capacity** is decoupled from reservations
- When you purchase Committed Use Discounts, it's not guaranteed you can launch an instance in the specified region or configuration.
- For a guarantee purchase a zonal reservation.

Google Cloud FinOps Fundamentals

VM Discounts

- ***Inferred instances***- for billing purposes, the same type of machine used in the same zone will be combined into a single charge (increases your discounts)
- Discount applied to custom machines as well
- Must be in same zone

<https://cloud.google.com/compute/docs/sustained-use-discounts>

<https://cloud.google.com/compute/docs/instances/signing-up-committed-use-discounts>

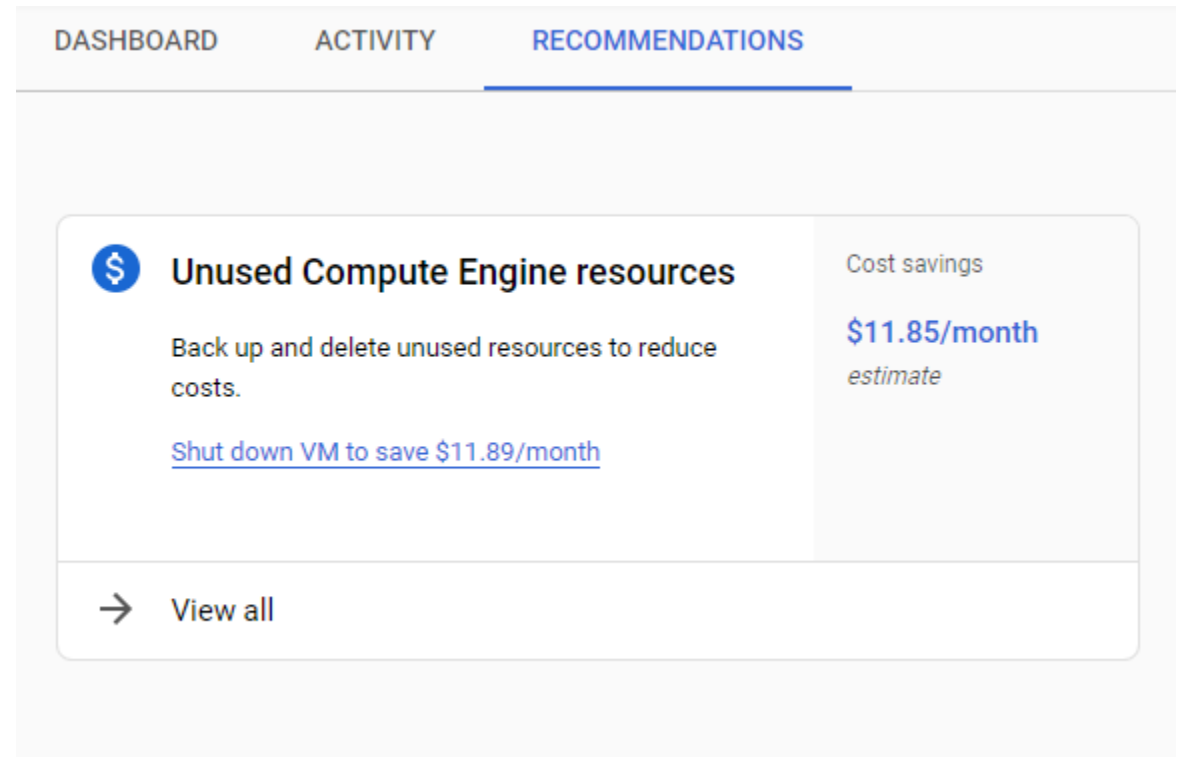
Active Assist Recommendations

Demo - Recommendations

Google Cloud FinOps Fundamentals

Recommendation Categories

- VM machine type recommender
- Committed use discount recommender
- Idle VM recommender
- Cloud SQL overprovisioned instance recommender



The screenshot displays the Google Cloud Recommendations dashboard. At the top, there are three tabs: 'DASHBOARD', 'ACTIVITY', and 'RECOMMENDATIONS', with 'RECOMMENDATIONS' being the active tab. Below the tabs, a recommendation card is shown for 'Unused Compute Engine resources'. The card features a blue circle with a white dollar sign icon. The title is 'Unused Compute Engine resources'. The description reads: 'Back up and delete unused resources to reduce costs.' Below this, there is a link: 'Shut down VM to save \$11.89/month'. To the right of the card, the 'Cost savings' are listed as '\$11.85/month estimate'. At the bottom of the card, there is a button with a right arrow and the text 'View all'.

Category	Recommendation	Cost Savings
Unused Compute Engine resources	Back up and delete unused resources to reduce costs. Shut down VM to save \$11.89/month	\$11.85/month estimate

Google Cloud FinOps Fundamentals

Cost Recommendations

←

Cost Recommendations

EXPORT

HISTORY

Recommendation Hub

INSTANCES

DISKS

IMAGES

IP ADDRESSES

\$

Review idle Compute Engine VM instances

Deleting instances that aren't used can help reduce costs.

Cost savings

\$11.85/month

Recommendations

DISMISS

Filter

Enter property name or value

?

Recommendation ↓	Resource name	Recommended action	Location	Refreshed
Shut down VM to save \$11.89/month	cloudfinopsdemo	Shut down	us-central1-a	Feb 13, 2022, 2:49:20 AM

Google Cloud FinOps Fundamentals



Finding Storage Waste

How to identify cost optimization opportunities with Cloud Storage

Google Cloud FinOps Fundamentals

Cloud Storage

- Correct Region
- Correct regional/zonal configurations
- Storage Class
- Object Level – Lifecycle Management

Google Cloud FinOps Fundamentals

Cloud Storage

- Standard: Access regularly, no retention minimum. This is "hot" data.
- Nearline: Access less than once a month, retain for more than a month.
- Coldline: Access less than once a quarter, retain for more than a quarter.
- Archive: Access less than once a year, retain for more than a year.

Google Cloud FinOps Fundamentals

Cloud Storage Retention Period

<https://cloud.google.com/blog/products/storage-data-transfer/how-to-save-on-google-cloud-storage-costs>

Access Frequency	Retention Period			
	<1 mo	1–3 mo	3–12 mo	>12 mo
	>12/yr	Standard	Standard	Standard
	4–12/yr	Standard	Nearline	Nearline
	1–4/yr	Standard	Nearline	Coldline
	<1/yr	Standard	Nearline	Archive

Google Cloud FinOps Fundamentals

Cloud Storage

- Storage Costs grow exponentially when unchecked.
- Setup Budget just for Cloud Storage

← Edit Budget

Storage Budget

Name *
Storage Budget

Scope

A budget enables you to track your actual spend against your planned spend.

Time range
Monthly

The month starts on the first of the month and reset at the beginning of each month.

A budget can be scoped to focus on a specific set of resources.

Projects
All projects (2)

Services

Filter |Type to filter

1 of 1938 selected

- ☐ Anthos
- ☐ Anthos Service Mesh
- ☒ BigQuery
- ☒ Cloud Logging
- ☐ Cloud SQL
- ☒ Cloud Storage
- ☐ Compute Engine
- ☐ Google Click to Deploy Anthos Sample Deployment on Google Cloud (Preview)

[SHOW 100 MORE OF 1938](#)

Cost trend

February 1, 2021 – February 28, 2022

\$250
\$200
\$150
\$100
\$50
\$0

Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb

● Actual cost

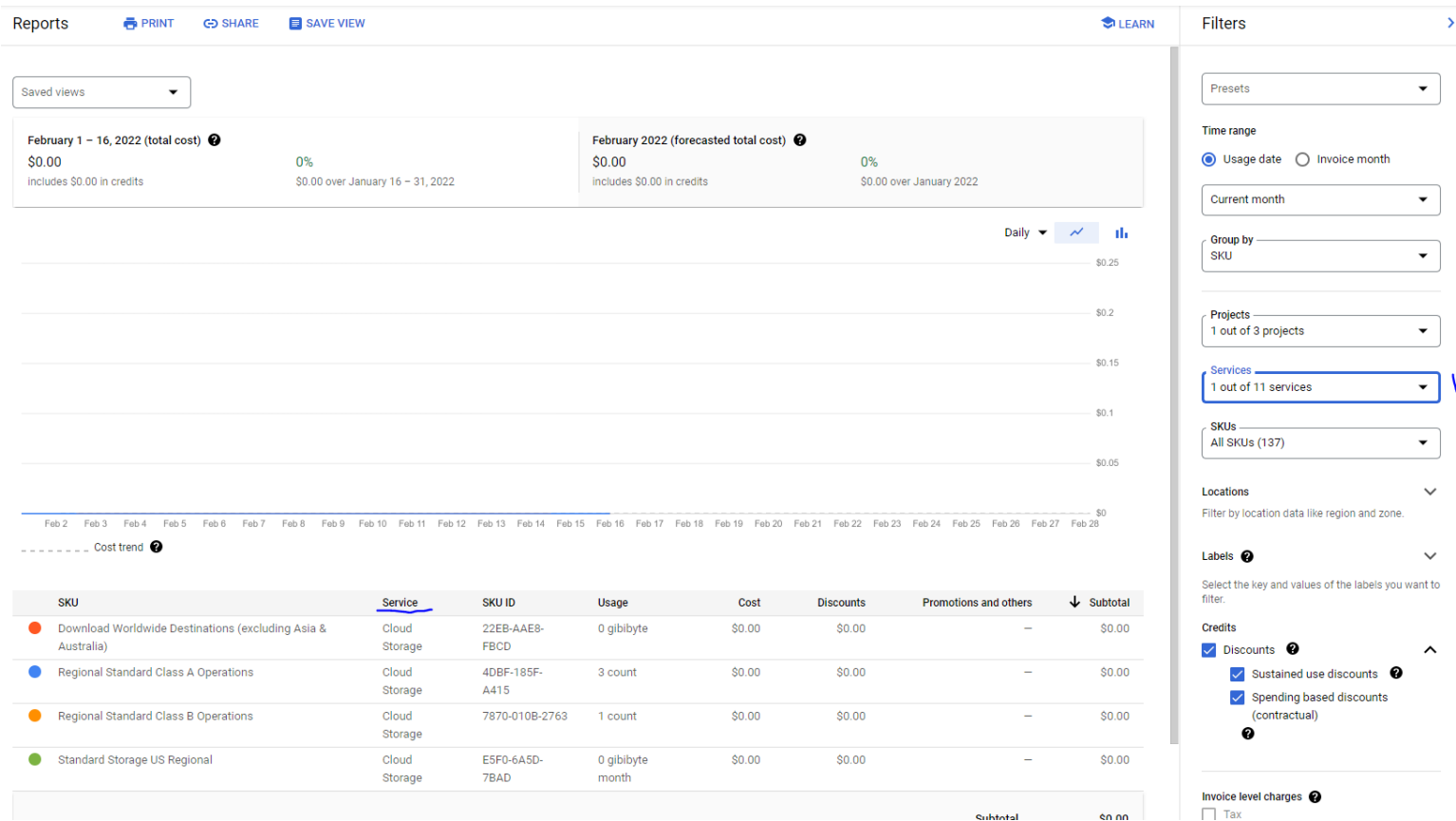
→ View report

SAVE CANCEL

Google Cloud FinOps Fundamentals

Cloud Storage

- Filter Billing Report



Finding Storage Waste

Demo - How to identify cost optimization opportunities with Cloud Storage

Google Cloud FinOps Fundamentals



Kubernetes Costs

Demo - Understanding the dynamics of Kubernetes costs

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Instance Discounts and Reservations

Demo – Compute Discounts/Reservations in GCP

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Kubernetes Engine Costs

Understanding the Charging model of GKE

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Cost Optimization Challenges with DevOps Environments

Understanding the dynamics to cost control with DevOps

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Optimization Challenges

DevOps consumes resources faster and unpredictably

Cost is not always a concern with DevOps engineers.
Cost vs Performance

Hard Coded Resources (IaC)

App Owner Resistance

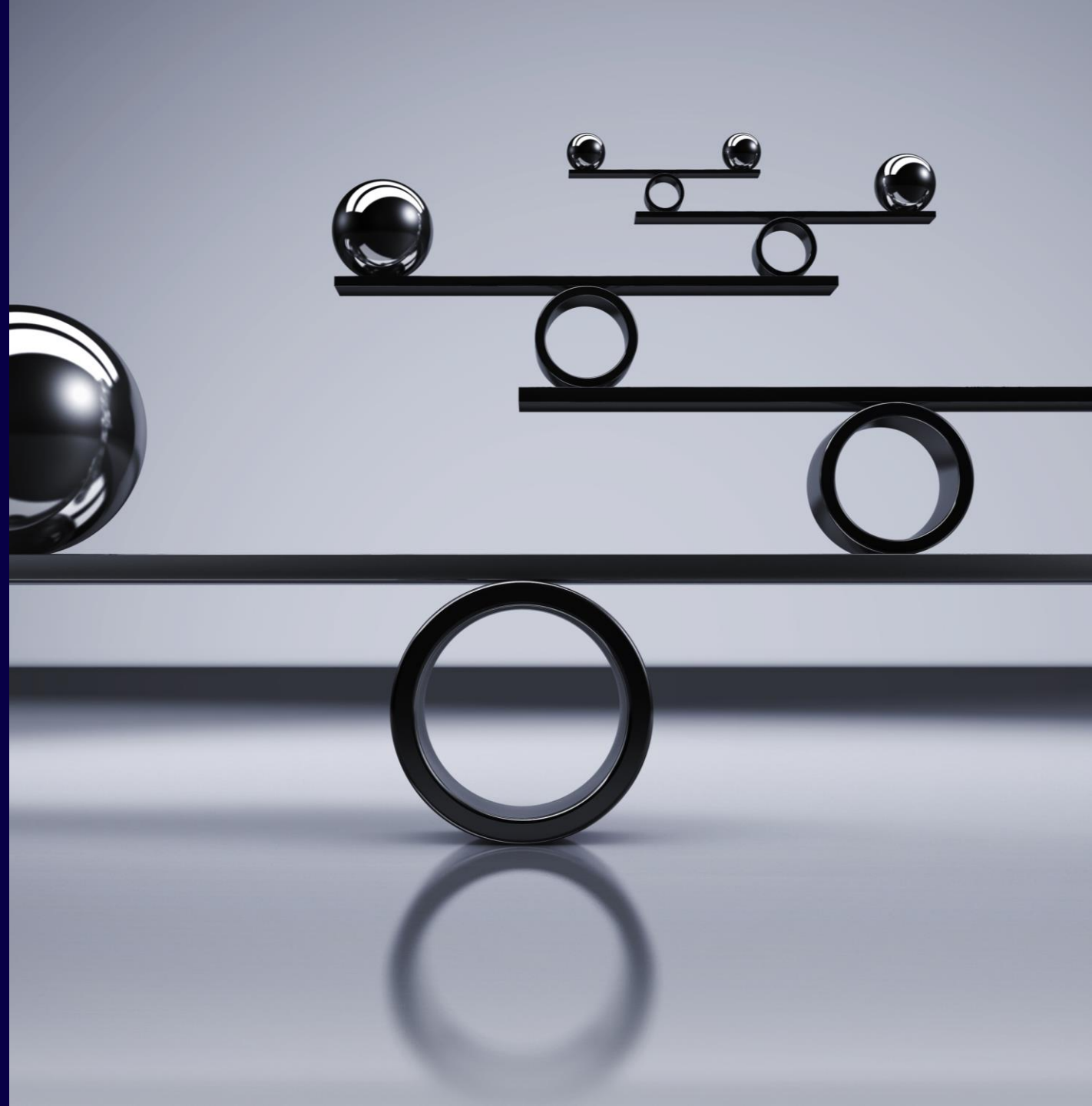
Traditional FinOps cost allocation doesn't work.

Multiple Container/Apps

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Optimization Challenges

- We now need to educate engineering about costs.
- A lot of the challenge comes from Kubernetes and how containers are deployed.
- Collaboration issues can hamper optimization
- Lack of Corporate direction(buy in and policy)



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Container Spending To Do List

- Setup cost allocation
- Visibility (Graphs)
- Setup QOS/Autoscaling
- Use Best Effort allocation
- Use Burstable allocation
- Understand Static vs Runtime costs
- Labeling(App,Env, Role) and Namespace (Segment)
- Rightsizing
- Discounts
- Kubernetes Cost Tools(Kubecost,Grafana)

Container Services

- Amazon Elastic Kubernetes Service (Amazon EKS)
- Google Kubernetes Engine (GKE)
- Azure Kubernetes Service (AKS)
- IBM Cloud Kubernetes Service (CKS)

Container Spending Example

- Billing Hierarchy: Organizations, folders, projects, normalizing them with cross-cloud concepts: Linked Accounts, Tags, Subscriptions, etc.
- Resources: Compute cores, RAM, GPU, TPU, Load Balancers, Persistent Disk, Custom Machines, Network Egress
- Namespaces: labeling specific, isolated containers
- Labels: Teams, cost centers, app names, environment, and more

<https://www.finops.org/projects/calculating-container-costs/>

BigQuery Pricing Model

Cost Management

BigQuery Cost Management

- BigQuery is a serverless, highly scalable, and cost-effective multi-cloud data warehouse designed for business agility.
- BigQuery's pricing and performance is priced differently than SQL.
- BigQuery on-demand (Higher Rate)
- BigQuery flat-rate discounts (Cost efficiency) for customers who prefer a stable cost for queries vs the on-demand *price per TB of data processed*.
- *BigQuery` does have some free operations.(Load, View, Copy, Export, etc)*

BigQuery

BigQuery pricing components:

- Analysis pricing: Process queries, user-defined functions, scripts, and certain data manipulation language (DML) and data definition language (DDL) statements that scan tables. (Billed per Second)
- Storage pricing: Store data loaded into BigQuery.

BigQuery

BigQuery slots:

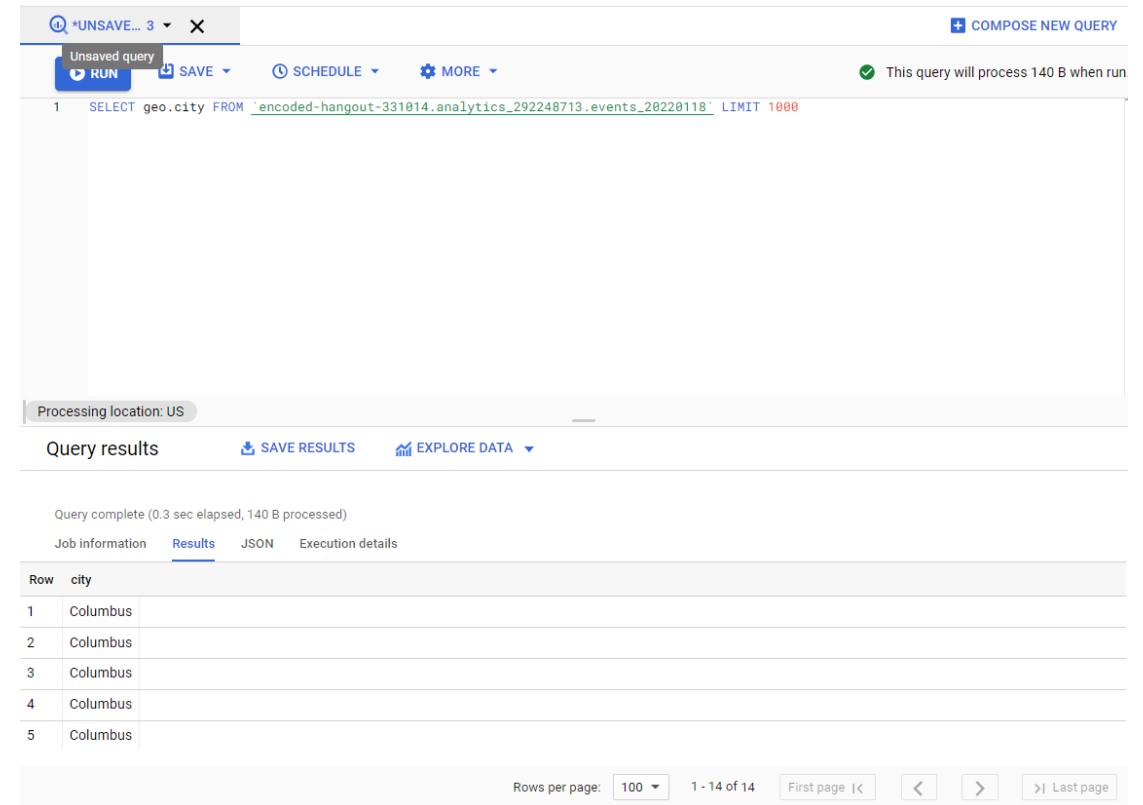
- A BigQuery slot is a unit of computational capacity used to execute SQL, DDL, and DML statements in BigQuery.
- Has a 100-slot minimum and is purchased in increments of 100 slots.
- On-Demand and Flat Rate Options

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BigQuery

BigQuery pricing components:

- BigQuery charges for certain other operations, including streaming inserts and using the BigQuery Storage API.



The screenshot displays the Google Cloud BigQuery console interface. At the top, there's a header with a tab labeled 'UNSAVE... 3' and a 'COMPOSE NEW QUERY' button. Below the header, a toolbar contains buttons for 'RUN', 'SAVE', 'SCHEDULE', and 'MORE'. A status message on the right indicates 'This query will process 140 B when run'. The main area shows a SQL query: `1 SELECT geo.city FROM `encoded-hangout-331014.analytics_292248713.events_20220118` LIMIT 1000`. Below the query, it states 'Processing location: US'. The 'Query results' section shows 'Query complete (0.3 sec elapsed, 140 B processed)' and tabs for 'Job information', 'Results', 'JSON', and 'Execution details'. The 'Results' tab is active, displaying a table with two columns: 'Row' and 'city'. The table contains five rows, all with the value 'Columbus'. At the bottom, there's a pagination bar showing 'Rows per page: 100', '1 - 14 of 14', and navigation buttons for 'First page', 'Previous', 'Next', and 'Last page'.

Row	city
1	Columbus
2	Columbus
3	Columbus
4	Columbus
5	Columbus

BigQuery

Demo - BigQuery Query

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Data Studio

Demo - BigQuery and Data Studio

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Visualizing your Data

Visualizing data and your options

Visualization Options

- Google Cloud Data Fusion, a cloud-native data integration service.
- BigQuery, a cloud data warehouse
- Google Data Studio, a relatively simple platform for reporting and visualization
- Google Cloud DataLab, a more robust analytics tool that lets data professionals explore, analyze, transform, and visualize data and build machine learning models
- Google Cloud DataPrep, a data service for exploring, cleaning, and preparing structured and unstructured data
- Google Cloud Dataflow, a platform for ingesting and processing real-time data

Cloud Operations

Monitoring and Logging

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Cloud Operations is a hybrid Monitoring, logging, and diagnostics for applications on Cloud Platform and AWS.

GCP Purchased Stackdriver and rebranded it to Google Stackdriver. (Now Cloud Operations)

Cloud Operations monitors the clouds service layers in a single SaaS solutions.

Native integration with Google Cloud data tools BigQuery, Cloud Pub/Sub, Cloud Storage, Cloud Datalab, and out-of-the-box integration with all your other application components.

Access from GCP Console

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Monitoring



Debugger



Logging



Trace



Error Reporting



Profiling

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Benefits

- Monitors multi cloud
- Identify trends and prevents issues
- Lowers Monitoring headaches
- Fix problems faster
- Reduces monitoring noise!
- Aids with Cloud Security
- Aids with Compliance



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Cloud Operations

Cloud Operations Account Options

- Create Operations account for project
- Create Operations account that monitors multiple projects

STACKDRIVER



Monitoring



Debug



Trace



Logging



Error Reporting



Profiler

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Cloud Operations

- Monitoring
- Defaults are intelligent and dynamic
- Health checks
- Metrics = Platform, system, application
 - >>>>Ingest Data Metrics, events and metadata
 - >>>>>Then provides insight thru dashboards, charts and alerts

Google Cloud FinOps Fundamentals

Cloud Operations



Cloud APIs



Cloud Deployment Manager



Cloud Endpoints



Debugger



Error Reporting



Logging



Monitoring



Stackdriver



Trace

- Metrics Examples

Compute Engine

- firewall/dropped_bytes_count
- firewall/dropped_packets_count
- instance/cpu/reserved_cores
- instance/cpu/usage_time
- instance/cpu/utilization
- instance/disk/read_bytes_count
- instance/disk/read_ops_count
- instance/disk/throttled_read_bytes_count
- instance/disk/throttled_read_ops_count

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Workspaces

- Workspace is a tool for monitoring resources contained in one or more GCP projects or AWS accounts.
- Each Workspace can have between 1 and 100 **monitored projects**, including one or more GCP projects and any number of AWS accounts.
- You can have as many Workspaces as you wish, but GCP projects and AWS accounts cannot be monitored by more than one Workspace.
- A Workspace contains the custom dashboards, alerting policies, uptime checks, notification channels, and group definitions that you use with your monitored projects.
- A Workspace can access metric data from its monitored projects, but the metric data and log entries remain in the individual projects.

Cloud Operations

Demo - Dashboards

Google Cloud FinOps Fundamentals



Account Billing

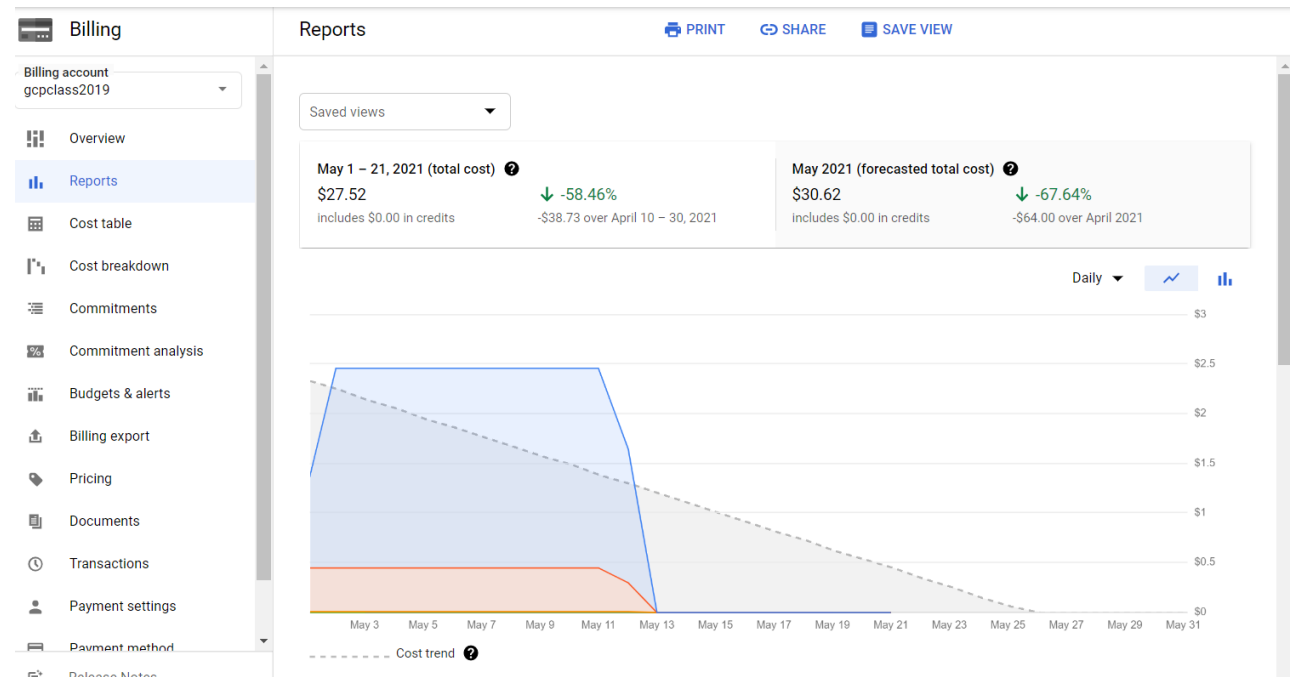
Understanding Billing and Discounts

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Billing Model

A Cloud Billing account is set up in Google Cloud and is used to define who pays for a given set of Google Cloud resources and Google Maps Platform APIs.

A Cloud Billing account is a cloud-level resource managed in the Cloud Console.



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Billing Model

Two types of billing accounts

1. Self Serve– Online - Payment instrument is a credit or debit card or ACH direct debit
2. Invoiced – Offline – Payment can be check or wire.

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Billing Model

Two types of payment profiles

1. Individual
2. Business

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Billing Model

Charging Cycles

Self-serve Cloud Billing accounts, your Google Cloud costs are charged automatically in one of two ways:

1. Monthly billing: Costs are charged on a regular monthly cycle.
2. Threshold billing: Costs are charged when your account has accrued a specific amount.

Demo – Budgets

Demo

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Demo – Budgets Notifications

Demo

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Demo – Quotas

Demo

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Cloud Language and Business Language

Understanding how Cloud Provider Naming correlate to business language

Providers and Service Names

- AWS
- Azure
- GCP



Comparing Service Provider Terms/Language

VM Consumption	Provider	Provider Name for Reserved Instance
Virtual Machine Reservation	AWS	Reservations/Savings Plans
Virtual Machine Reservation	Azure	Reservations
Virtual Machine Reservation	GCP	Committed Use Discounts/Flat Rate Plan

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Comparing Service Provider Terms/Language

Compare AWS and Azure services to Google Cloud 

[Send feedback](#)

Last updated: August 31, 2021

This table lists [generally available](#) Google Cloud services and maps them to similar offerings in Amazon Web Services (AWS) and Microsoft Azure. You can filter the table with keywords, such as a service type, capability, or product name. We welcome your [feedback](#) to help us keep this information up to date!

Filter by entering a product name or characteristic.

Service category ▾	Service type	Google Cloud product	Google Cloud product description	AWS offering	Azure offering
App Modernization	CI/CD	Cloud Build	Build, test, and deploy on Google Cloud serverless CI/CD platform	AWS CodeBuild, AWS CodeDeploy, AWS CodePipeline	Azure DevOps, Github Enterprise
App Modernization	Multi-cloud	Anthos	Anthos is a managed application platform that extends Google Cloud services and engineering practices to your environments so you can modernize applications faster and establish	AWS Outposts	App Modernization with Microsoft Azure, Azure Arc

<https://cloud.google.com/free/docs/aws-azure-gcp-service-comparison>

Whiteboard – Putting it all Together

Summary

Google Cloud FinOps Fundamentals

Putting it all Together



Section Summary

Section 3 : Managing Service Costs



3

Section

Section Summary

- Active Assist has Recommendation Categories such as VM machine type recommender and Committed use discount recommender.
- BigQuery has two distinct pricing models which are on demand and flat-rate.
- BigQuery is a managed data warehouse that we can export data to and then visualize with Cloud Data Studio
- Forecasting can be done by viewing billing and we can use budgets to notify us of billing thresholds that have been reached.
- Compute Engine has several discount models. VM vCPU and memory usage for each of these categories can receive one of the following discounts which are Sustained use discounts, Committed use discounts and discounts for preemptible VM instances
- Cloud Operations is a monitoring tool we can use to monitor resources performance and consumption. We can create a custom dashboard to alleviate searching.

Section Review Questions

Section 3 : Managing Service Costs



Review Questions

BigQuery has two distinct pricing models. What are they? (Select Two)

- On Demand
- Flat Rate
- Spot
- Reserved

Review Questions

BigQuery has two distinct pricing models. What are they? (Select Two)

- On Demand
- Flat Rate
- Spot
- Reserved

Review Questions

Cloud Storage provides four distinct ‘classes’ of storage performance and availability? What are they? (Select Four)

- Standard
- Nearline
- Glacier
- Spot
- Coldline
- Archive

Review Questions

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Review Questions

Google Cloud provides what is a preemptible instance. What are the features of this instance type? (Select Three)

- Lives at most 24 hours
- Can be pre-empted with a 30 second notification via API
- Discounted significantly
- Higher performance
- Lower performance

Review Questions

Google Cloud provides what is a preemptible instance. What are the features of this instance type? (Select Three)

- Lives at most 24 hours
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Review Questions

BigQuery is a managed data warehouse that we can export data to and then visualize with what data service directly from BigQuery? (Select One)

- Cloud DataLab
- Cloud Storage
- Cloud Data Fusion
- Cloud Data Studio

Review Questions

BigQuery is a managed data warehouse that we can export data to and then visualize with what data service directly from BigQuery? (Select One)

- Cloud DataLab
- Cloud Storage
- Cloud Data Fusion
- **Cloud Data Studio**

Additional Resources

Resources to get started with.

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Resources

- TheGCPGurus.com
- [TechCommanders](#)
- [GitHub](#)
- [Google Cloud Platform Documentation](#)

Interview Coaching and Enablement

Get ready to interview and get the offer.

Cloud InterviewACE Program Overview



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Resources

- Cloud InterviewACE
- Hybrid Training and Coaching program



<https://www.techcommanders.com/cloudinterviewace>

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Resources

- Book Release February 4th, 2022
- <https://amzn.to/3EPzmRV>



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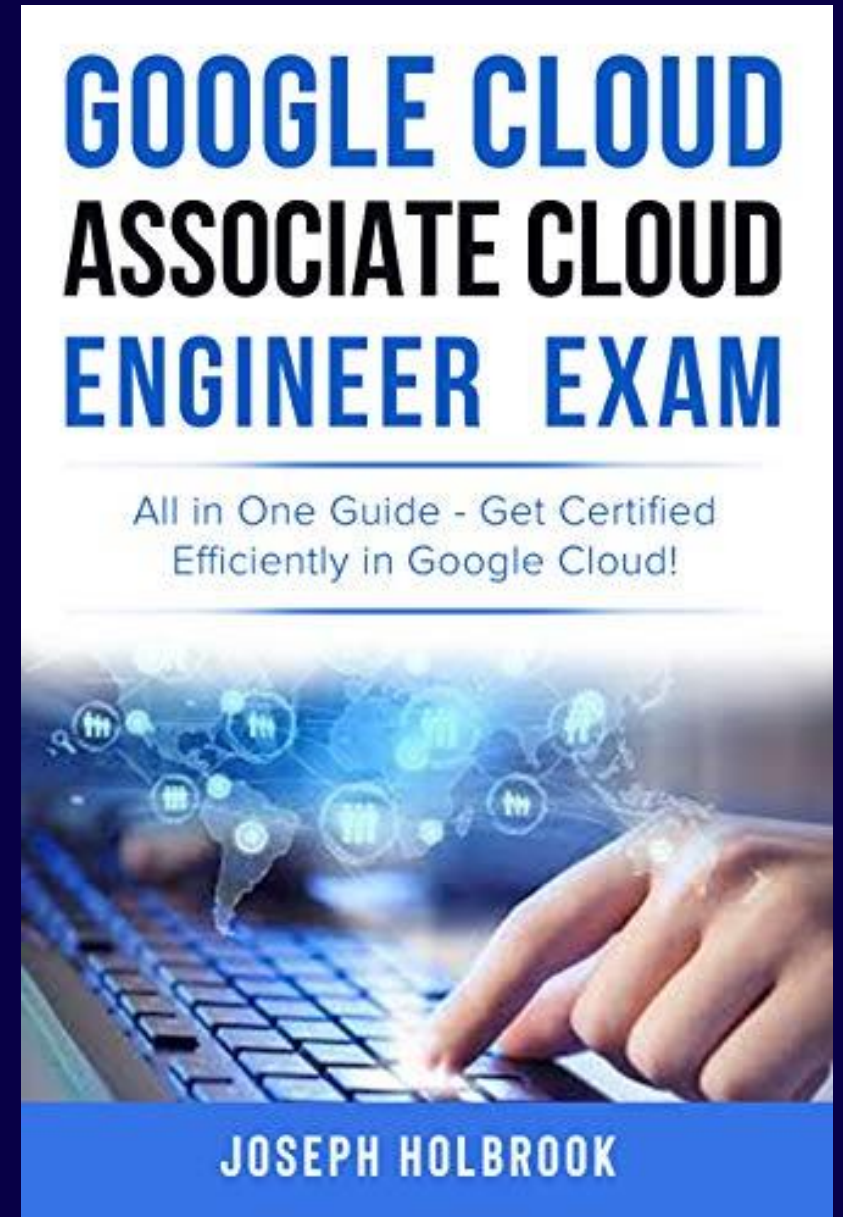


JOSEPH HOLBROOK

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Resources

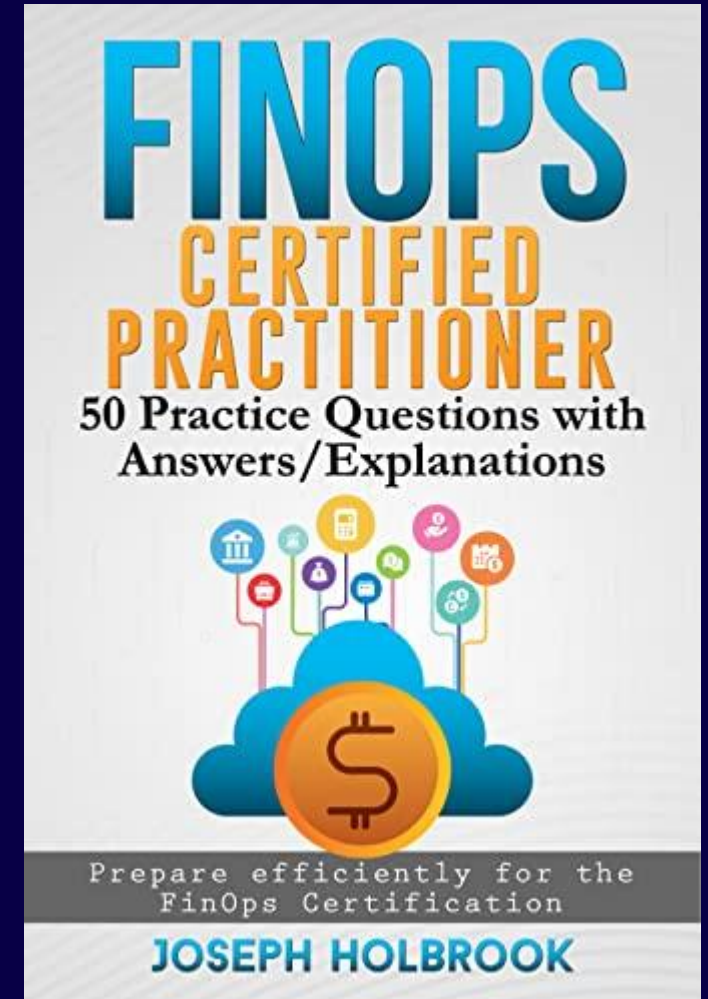
- Available now
- <https://amzn.to/3eJHlzR>



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Resources

- Available now
- <https://amzn.to/3gSdTOv>



Course Closeout

Thank you for Joining

Course Closeout

Thank you for joining the course.

I wish you much success on the exam and in your life!

Please reach out if I can be of assistance.



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